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VMware vSphere Resource Management Essentials

Optimum solutions to help you manage your VMware vSphere resources effectively

Jonathan Frappier

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First published: February 2014

Production Reference: 1130214

Published by Packt Publishing Ltd.
Livery Place
35 Livery Street
Birmingham B3 2PB, UK.

ISBN 978-1-78217-046-4

www.packtpub.com

Cover Image by Jeremy Segal (info@jsegalphoto.com)

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Jonathan Frappier

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About the Author

Jonathan Frappier is a hands-on technology professional with over 15 years of experience in VMware-virtualized environments, focusing on system interoperability. He has specialization at the intersection of system administration, virtualization, security, cloud computing, and social enterprise collaboration.

He had not touched a computer until high school but then quickly found his passion. Jonathan holds a Bachelor's degree in Computer Science from Newbury College and a Master's degree in Computer Information Systems from Boston University, which he completed while working full time. He holds VMware certifications, including VCAP5-DCD, VCP5-DCV, VCA-Cloud, DCV, and WM.

Jonathan has worked in enterprises and start-ups throughout his career and has become a self-defined jack of all trades, but he is most passionate about virtualization and its community, and was honored as a vExpert 2013 for his contributions.

You can find Jonathan on Twitter `@jfrappier`, and on his blog at `www.virtxpert.com`, as well as at almost every Virtualization Technology User Group (VTUG) meet. He also supports the #vBrownBag podcast at `professionalvmware.com`.

First and foremost, I'd like to thank my wife for putting up with me for all these years and my daughter who will have to put up with me for many more. I am thankful to my parents for supporting me even when I had no idea why I was going to college. I would also not be here if it were not for my grandparents, who I was fortunate to know well in my adult life, and in loving memory of my grandfather, Norman L'Heurex, and Pepere Lionel Frappier. Just as much love is due to my friends Jim, Jeremy, Manny, Bob, and Igor.

I'd also like to thank the following people from the virtualization community: Matt, Sean, Luigi, and others—the "Nerdherd". I'm not sure I'd be doing this if it weren't for that dinner at Ichigo Ichie. Also, a big thank you to the #vBrownBag crew for giving me the opportunity to give back through this incredible organization; to all the folks out there blogging, sharing, and helping others; and the folks at VMware who support the community. Your contribution inspires me.

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About the Reviewers

Angelo Luciani is an IT Professional in the financial industry who specializes in VMware virtualization and systems management. He is also the Toronto VMware User Group (VMUG) leader and enjoys contributing to the virtualization community. You can connect with Angelo at `www.twitter.com/AngeloLuciani` or head over to his blog at `www.virtuwise.com`.

Mario Russo has worked as an IT Architect, a Senior Technical VMware Trainer, and in the presales department. He has also worked on VMware technology since 2004.

In 2005, he worked for IBM on the first large project Consolidation for Telecom Italia on the Virtual VMware ESX 2.5.1 platform in Italy with the Physical to Virtual (P2V) tool. In 2007, he conducted a drafting course and training for BancoPosta, Italy, and project disaster and recovery (DR Open) for IBM and EMC. In 2008, he worked for the project Speed Up Consolidation BNP and the migration of P2V on the VI3 infrastructure at BNP Cardif Insurance.

He is a VCI certified instructor level 2s of VMware and is certified in VCAP5-DCA.

He is the owner of Business to Virtual, which specializes in virtualization solutions.

He was also the technical reviewer of many books, including *Implementing VMware Horizon View 5.2* by *Jason Ventresco*, *Implementing VMware vCenter Server* by *Konstantin Kuminsky*, *Troubleshooting vSphere Storage* by *Mike Preston*, *VMware Horizon View 5.3 Design Patterns and Best Practices* by *Jason Ventresco*, all published by *Packt Publishing*.

I would like to thank my wife, Lina, and my daughter, Gaia. They're my strength.

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Akmal Khaleeq Waheed is a solutions architect working on virtualization technologies. VMware being his primary expertise, he keeps an eye on competitive virtualization technologies such as MS, Citrix, and Red Hat. He has previously worked with Enterprise Server at Hewlett Packard and virtualization at VMware, Inc. He is VCA, VCP, VCAP-DCA, and DCD certified and the first winner of Virtual Design Master 2013-VDM001, first ever IT reality competition organized by the VMware community.

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Preface

VMware vSphere Resource Management Essentials provides readers with a high-level understanding of the various components, methodologies, and best practices for maintaining and managing resources in a virtual environment.

Readers will begin the book by going through an explanation of the requirements for ESXi and the foundation for VMware vSphere. Also, this book will provide readers with an understanding of how resources are supplied and the features that enable resource and virtual machine availability.

With an understanding of the requirements to build and run your environment, you will then move into understanding how ESXi manages resources such as CPU, memory, disk, and networks for multiple virtual machines and ensures there is resource availability.

With VMs built and resources assigned, readers will get to know the advanced features as well as the monitoring and automation tools included to make your VMware vSphere environment more efficient.

What this book covers

Chapter 1, *Understanding vSphere System Requirements*, provides specific resource requirements for installing ESXi and vCenter, as well as providing links to online resources such as the VMware HCL.

Chapter 2, *Assigning Resources to VMs*, covers how virtual machines use physical resources provided by ESXi hosts, and provides various techniques that ESXi uses to manage the allocation of physical resources.

Chapter 3, *Advanced Resource Management Features*, provides an overview of the various tools and features licensed with VMware vSphere to increase resource utilization and availability.

Chapter 4, *Automation and Monitoring Options*, takes a look at the two main automation tools available with VMware vSphere, PowerCLI, and vCenter Orchestrator; it also covers monitoring solutions built into vCenter and vCenter Operations Manager.

What you need for this book

This book can be read without any additional resources; however, we recommend that you have access to some physical lab resources to install ESXi and vCenter on, and at least a trial of the VMware vCenter Operations Management Suite available at `www.vmware.com/go/try-vcenter-ops`.

A reasonable lab setup can be one on a single physical host with at least a dual or quad core processor and 8 GB of RAM (4 GB for vCenter and 2 GB for a Windows Server VM to run Active Directory).

Who this book is for

If you are a vSphere administrator who wants to understand new features or an administrator aspiring to start your virtualization journey with VMware vSphere and want to learn how to manage your resources, this book is for you.

Conventions

In this book, you will find a number of styles of text that distinguish between different kinds of information. Here are some examples of these styles, and an explanation of their meaning.

Code words in text, database table names, folder names, filenames, file extensions, pathnames, dummy URLs, user input, and Twitter handles are shown as follows: "To see the information, you just need to run the `sar` command."

Any command-line input or output is written as follows:

```
Set-VM -VM <name_of_vm> -MemoryMB 8192
```

New terms and **important words** are shown in bold. Words that you see on the screen, in menus or dialog boxes for example, appear in the text like this: "Once the **Performance Monitor** window opens, expand **Monitoring Tools** and click on **Performance Monitor**."

Warnings or important notes appear in a box like this.

Tips and tricks appear like this.

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1
Understanding vSphere System Requirements

VMware vSphere allows organizations to achieve extremely high levels of resource utilization for both new and existing physical servers by allowing administrators to easily virtualize and consolidate existing servers. By consolidating multiple servers and applications onto a single physical server, organizations can maximize their **Total Cost of Ownership** (**TCO**) by using more of the resources provided by the physical server. However, like any other system, there are certain requirements vSphere needs to fulfill in order to function properly. In addition to the system requirements for various vSphere components that make up a working environment, it also brings back the lost art of true infrastructure design. This ensures that you understand the requirements of the systems and applications you will be virtualizing so that they can operate according to your organization's **Service Level Agreements** (**SLA**).

In this chapter, we will review the requirements for installing various components of vSphere, including ESXi and vCenter. Also, we'll learn about the tools available to validate your hardware and look at the basics of determining resource utilization for your existing servers and applications.

The topics we'll cover in this chapter are as follows:

- Understanding ESXi system requirements and general best practices
- Understanding vCenter system requirements and options
- Understanding the features of vSphere
- Reviewing the basic components of a vSphere environment
- Understanding configuration maximums
- Determining resource utilization requirements

ESXi system requirements

VMware vSphere Hypervisor, also known as ESXi, is the foundation on which you will build your vSphere environment. A hypervisor is simply software that manages and assigns physical resources to virtual systems. Hypervisors are classified into two types: type 1 and type 2. Type 1 hypervisors are installed onto bare metal hardware such as ESXi. Type 2 hypervisors are installed on top of an existing operating system, for example, a VMware Workstation, which is installed on Windows or Linux, or VMware Fusion, installed on OSX.

ESXi is installed on your physical server to a hard drive, USB flash drive, or a built-in chip directly from manufactures such as Hewlett Packard. Alternatively, it can be directly installed into the memory using an advanced feature called **auto deploy**. By using vSphere, applications that may have otherwise not utilized all of the resources in a physical server will now be shared among multiple **Virtual Servers** (**VM**s) and the applications running on them. VMware previously released other hypervisors, for example, VMware server that was installed on top of Windows systems that allowed you to create virtual servers. At the time of writing this, VMware is committed to using ESXi as its type 1 hypervisor.

As you have most likely become accustomed to other operating systems and applications, there are specific system requirements based on the version of ESXi that you are installing. The VMware Knowledge Base maintains up-to-date information not only about the system requirements for the version of ESXi that you are running, but also about known problems and steps for troubleshooting error messages you may encounter.

VMware KB 2052329 provides a list of the minimum requirements for installing ESXi 5.5 available at `http://kb.vmware.com/kb/2052329`.

If you have an older version of ESXi that you need to continue to support, you can find the requirements in VMware KB 1003661 available at `http://kb.vmware.com/kb/1003661`.

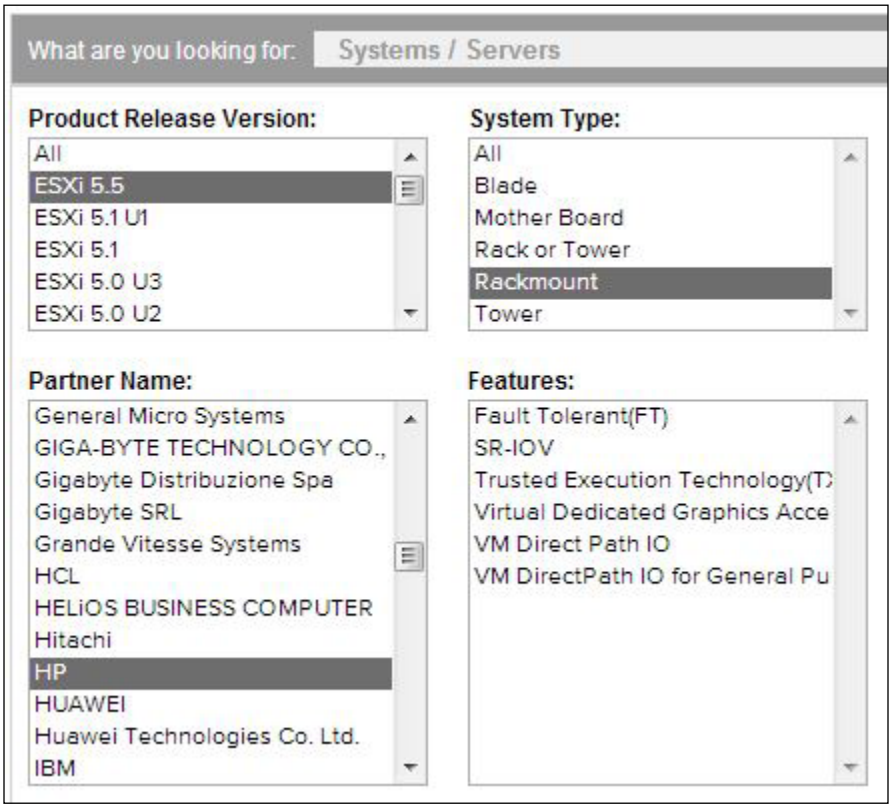
Let's review some of the resource requirements for ESXi 5.5 as follows:

- A 64-bit x86 processor with at least two cores, such as an Intel Xeon E5-2640 or AMD Opteron 6320, which supports hardware virtualization (Intel VT-x or AMD RVI).
- A minimum of 4 GB RAM to install ESXi is required. However, this is truly a minimum requirement as the consolidation ratios and application requirements you are targeting will ultimately drive the amount of RAM in your servers. We will discuss consolidation ratios later in this chapter.

- At least one gigabit network adapter.
- A supported storage controller and a device with minimum of 1 GB of space to install ESXi and 5.2 GB to create the scratch partition.

Additionally, you will want to confirm that all of your hardware is supported by visiting the *VMware Compatibility Guide* available at `http://vmware.com/go/hcl`.

Now, if you are thinking, "Oh great, I need to go through all the hardware in my server to make sure it works!", be rest assured that VMware has a vast partner network. Therefore, validating servers from vendors such as HP, Dell, and Cisco can be done with just a few clicks, as seen in the following screenshot:



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In the preceding screenshot, we searched for **Rackmount** servers that are compatible with **ESXi 5.5** from HP. Selecting these three options and clicking on the **Update and View Results** button give us a list of servers that match our search, as shown in the following screenshot:

Search Results: Your search for "Systems / Servers" returned **39 results**.

Partner Name	Model	CPU Series	Supported Releases	
HP	ProLiant DL580 G7	Intel Xeon E7-4800 Series	ESX	4.1 U3
			ESXi Installable	4.1 U3
			ESXi Embedded	4.1 U3
			ESXi	5.5
HP	ProLiant DL580 G7	Intel Xeon E7-8800 Series	ESX	4.1 U3
			ESXi Installable	4.1 U3
			ESXi Embedded	4.1 U3
			ESXi	5.5

Of course, if you build your own servers, this process may be a little more manual since you will have to verify each component.

One last site you should bookmark and get in the habit of visiting is the VMware Community site. The Community site is filled with people passionate about virtualization, supporting and giving back to others in the virtualization community. It's a great place to ask questions and see what others have been experiencing. For example, maybe you are trying to decide between two quad-port NICs for your server, which are both in the compatibility guide, but the chipset on one is more prone to errors than the other. If that were the case, I bet there would be a discussion on this! You can find the VMware Community site at `https://communities.vmware.com/welcome`.

Now that we know the requirements and how to find the requirements for your specific environment, installing ESXi on a server, starting to build virtual machines, or even converting current physical machines to virtual machines is within your reach. If you think consolidating multiple physical servers onto a single ESXi host is amazing, wait until you see what managing ESXi hosts with vCenter can do.

vCenter components

While ESXi is the foundation on which we build our virtual environment, the amazing features that allow greater resource utilization of our physical servers comes from joining multiple ESXi hosts together and managing them with VMware vCenter. vCenter itself comes in two different versions, standard and foundation. The **foundation** edition is designed for small deployments, supporting only three hosts, and is generally packaged into *essentials kits*. The **standard** edition is required for anything beyond three managed hosts.

This probably won't be the last time I say this, but just as any other application, you need to plan your vCenter deployment properly to ensure it can support your environment. VMware provides two basic options for deploying vCenter: either installed as an application on top of Microsoft Windows or as a prepackaged Linux virtual appliance. The virtual appliance is not new to vSphere 5.5, but VMware has expanded capabilities of the appliance in vSphere 5.5 to support up to 100 ESXi hosts or 3,000 virtual machines using the built-in PostgreSQL database or up to 1000 hosts or 10,000 virtual machines with an external Oracle database. You can find more information in the *vSphere Configuration Maximums* guide, which we will review shortly. It is available at `http://kb.vmware.com/kb/2052334`.

As of vSphere 5.1, VMware separated many functions that were once bundled together into separate services, allowing you for having greater control (and greater complexity) of your vCenter deployment. The main components you need to be aware of are as follows:

- vCenter
- vCenter Single Sign-On
- vCenter Inventory Service
- vSphere Web Client

Covering each of these components is beyond the scope of this book. For a more in-depth look at each component, check out *Implementing VMware vCenter Server*, *Konstantin Kuminsky* from *Packt Publishing* or *Mastering VMware vSphere 5.5 from Sybex*, *Nick Marshall* and *Scott Lowe* and several other authors who are prominent members of the VMware community.

vCenter also requires a database server to store configuration information, collect statistics about the environment, and perform various tasks to maintain a healthy database. The database platform you require (yes, I'm going to say it again) needs to be planned properly in order to support your requirements. vCenter can be installed on the same server as your database, but I would treat defining system requirements as if they were going to be on separate servers.

VMware considers a *small* deployment for vCenter up to 100 hosts or 1000 VMs; for many organizations, this is probably adequate but can support much more. The resources required for vCenter to run efficiently are based on your host or VM inventory.

A **Simple Install,** which is a streamlined installation of vCenter, vCenter SSO, vCenter Inventory Service, and the web client, has the following requirements:

- 2x 64-bit processor cores (such as ESXi; these can be either Intel or AMD) with a speed of 2 GHz
- 12 GB of memory
- 40 to 60 GB of disk space
- 64-bit database server

Again, it's important to emphasize that this is in addition to your database server requirements. While the database server requirements, installation, and configuration are beyond the scope of this book, let's take a quick look at the system requirements for a database server.

Microsoft recommends a 2 GHz or a faster processor of 4 GB of RAM along with 3.6 GB of free disk space for various SQL Server components. When added to the vCenter server requirements, you have what may be one of the larger servers in your environment. You should consult your DBA team, or if it is a one-man show, refer to Google and the SQL documentation online at `http://technet.microsoft.com/en-us/library/ms143506(v=sql.105).aspx`.

Understanding vSphere features

So we know that vCenter is available in two different versions, but what is really important is the features that vCenter unlocks. VMware is constantly innovating and adding new features, so it's best to visit the VMware website for the latest version and features available.

There are currently three editions of vSphere: Standard, Enterprise, and Enterprise Plus. Understanding what features are available in each edition is the most important step in building your virtualization plans so that you can purchase the edition that meets your requirement. Let's look at the three versions and what is available in each one of them. When you get to in *Chapter 3*, *Advanced Resource Management Features*, you'll notice that these line up pretty closely, so if you are unfamiliar with a feature mentioned in the following table, we will cover it shortly.

Feature	**Standard**	**Enterprise**	**Enterprise Plus**
vMotion	Yes	Yes	Yes
Storage vMotion	Yes	Yes	Yes
High Availability (HA)	Yes	Yes	Yes
Fault Tolerance (FT)	Yes	Yes	Yes
Hot Add	Yes	Yes	Yes
App HA		Yes	Yes
Distributed Resource Scheduler (DRS)		Yes	Yes
Storage DRS			Yes
Flash Read Cache			Yes
Virtual Distributed Switch (vDS)			Yes
Network I/O control			Yes
Storage I/O control			Yes
Host Profiles and auto deploy			Yes
VSAN	Licensed separately		

I added VSAN to this list and the book even though it is scheduled to be a separate product (VSAN is currently still in beta at the time of writing this); it is one of the most exciting pieces of technologies since virtualization.

As a VMware administrator, many of us would select Enterprise Plus for our licenses, but as always, cost is going to be a factor. As we review each of the features and their ability to help manage resources, you will be able to make a more informed decision on which version is right for your organization.

Topology basics

Now that you are familiar with the system requirements to get ESXi and vCenter installed, and have a list of the features available in different versions, it's time to look at how a typical vSphere deployment is organized. Different features of vSphere are configured and made available at various levels, so understanding each feature is critical. A basic vCenter environment will contain a data center, and within the data center, you create clusters. Hosts are placed into a cluster and then VMs are created on a host.

Understanding vSphere data center

The topology of a vSphere environment is very similar to what you might see from a physical perspective. After installing ESXi and vCenter, one of the first items you will be asked to create is a data center. A data center is, in its simplest form, the container (or a storage area if we are relating it to a physical environment) where your resources are stored. Since we live in a virtual world, your vCenter data center is not tied to one physical location. In fact, if you really wanted, you could create several vCenter data centers even if all the servers were in the same physical location, but there is not much need for that. It's common to mirror your vCenter data centers to physical locations of your ESXi hosts. For example, if you have a physical data center in Boston and Atlanta, you might create two data center objects in vCenter, one for Boston and one for Atlanta, which contain the physical resources located at each data center. Once your data center is created, you could move on to add hosts and leveraging features such as vMotion; however, you would be missing out on one of the critical areas as it relates to managing resources—vCenter clusters.

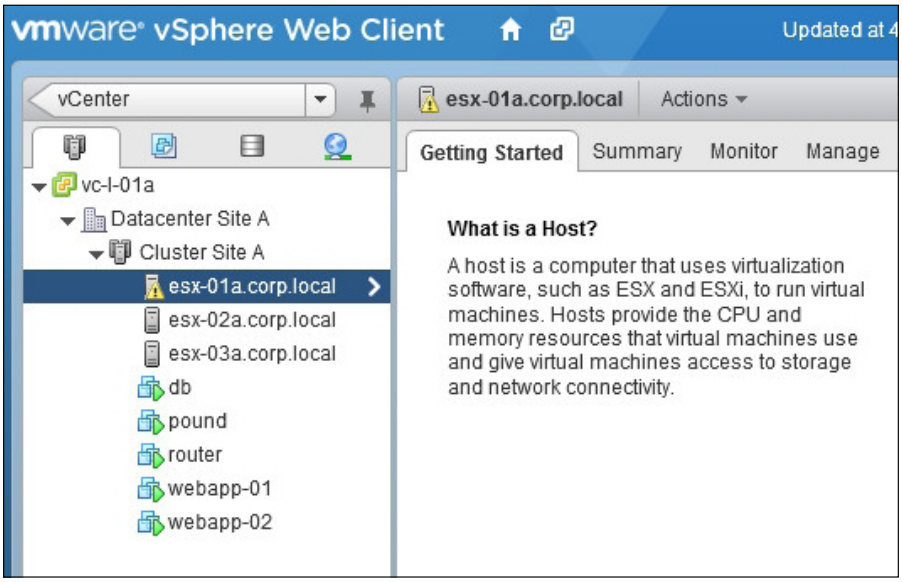
Familiarizing yourself with a vSphere cluster

Clusters are created within data centers and again, because this is all virtual, you could create as many clusters as you like. Clusters are a collection of ESXi hosts that can share resources amongst each other. In fact, two of the most important resource and availability features vSphere offers are enabled at the cluster level: **High Availability** (**HA**) and **Distributed Resource Scheduler** (**DRS**). There are several other settings configured at the cluster level that we will cover in *Chapter 3*, *Advanced Resource Management Features*.

What is a vSphere host?

As mentioned in the previous section, within a cluster, we add ESXi hosts. There is quite a bit of configuration done at a host level, but without a host, we cannot virtualize any servers or desktops. As a quick aside, much of what we talk about in this book relates to server virtualization, but both ESXi and vCenter also provide the necessary infrastructure for virtual desktops. Yes, you can virtualize your desktops as well! If you are interested in learning more about **Virtual Desktop Infrastructure** (**VDI**), check out *Implementing VMware Horizon View 5.2, Jason Ventresco, Packt Publishing*.

Now that we have our hosts in a cluster and our cluster in a data center, we can start creating VMs or even use a tool such as VMware Converter to take an existing physical server and convert it into a virtual server. Here is what a basic vSphere environment might look like:



Using the preceding screenshot as a reference, we have a vCenter Server called **vc-l-01a** and a data center called **Datacenter Site A**. Within this data center, we have a cluster called **Cluster Site A**, and within the cluster, we have three ESXi hosts; on the hosts, there are five VMs.

Remembering configuration maximums

Up until now, we have focused on the minimum requirements to run ESXi and vCenter; however, it is important to remember that there are actually maximum supported configuration settings as well. For example, while we have seen that vCenter, with an appropriate amount of resources, can manage up to 400 hosts or 4,000 VMs, a cluster can only contain 32 ESXi hosts. In this scenario, if you had 400 hosts, you would have to create 13 clusters within your data center (or data centers) to support all of those hosts. Depending on your organization, you may also have deployed multiple vCenter servers to manage all of those hosts.

All of the various components that make up a vSphere environment have various configuration maximum settings. Depending on your physical to virtual consolidation ratio, you could find yourself in a scenario where you were within the limits of one configuration maximum setting but are constrained by another. For example, you can configure up to 10 virtual NICs per virtual machine but can only have 1016 active ports per host. If you were able to run 102 VMs on a single physical host and each VM had 10 active virtual NICs, then you would need 1020 active ports, which is above the configuration maximum of 1016 per host. This scenario might be a bit extreme, but it's used to point out that just because you comply with the maximum configuration settings for one resource, it does not mean it would work with another.

Let's take a look at some of the more common configuration maximum settings.

Virtual machine maximums (per VM)

Every component within the hierarchy of a vSphere environment has certain maximum settings. Here are some common VM configuration maximums:

- 64 virtual CPUs
- 1 TB of RAM
- 62 TB of virtual disk size
- 10 virtual NICs
- 20 USB devices

ESXi host maximums (per host)

Like the VM configuration maximum settings mentioned in the previous section, ESXi hosts also have several maximum settings you need to be aware of when you need to design and configure your host, mentioned as follows:

- 320 logical CPUs
- 512 VMs
- 4096 virtual CPUs (total number of assigned virtual CPUs across all VMs)
- 32 virtual CPUs per physical core
- 4 TB of RAM
- 1 VSWP swap file per virtual machine
- 256 iSCSI, FC LUNs, or NFS mounts
- 64 TB of datastore volume size
- 256 volumes per host
- 24 e1000e 1 GB ethernet ports (Intel PCI-e)
- 8 bnx2x 10 GB ethernet ports (Broadcom)
- Eight 10 GB and four 1 GB ports (combined)

Cluster maximums

Cluster maximums are one of the more likely areas you will reach when you use configuration maximum settings during your design and implementation. Some of them are mentioned as follows:

- 32 hosts
- 4000 virtual machines
- 512 virtual machines per host
- 1600 resource pools per host

Like anything else, configuration maximums tend to change from version to version, so make sure you check what the configuration maximums are for your version of vSphere. Configuration maximums for vSphere 5.5 can be found at `http://www.vmware.com/pdf/vsphere5/r55/vsphere-55-configuration-maximums.pdf`.

Determining resource utilization requirements

Until now, we have focused on the minimum resource requirements to run ESXi and vCenter, but how do you determine the needs specific to your environment? Before virtualization, in many cases, the physical servers you were buying were so much more powerful than what many needed. This is why sizing became somewhat of a lost art.

For those hoping to find a magical catch-all formula that will work in every scenario, you'll have to keep looking. Remember every environment is unique, and even where similarities may arise, the use case your organization has, will most likely be different from another organization. Beyond your specific VM resource requirements, the hosts you are installing ESXi on will also vary; the hardware available to you will affect your consolidation ratio (the number of virtual machines you can fit on a single host). For example, if you have 10 servers that you want to virtualize, and you have determined each requires 4 GB of RAM, you might easily virtualize those 10 servers on a host with 48 GB of memory. However, if your host only has 16 GB of memory, you may need two or three hosts in order to achieve the required performance.

Another important aspect to consider is when to collect resource utilization statistics about your servers. Think about the requirements you have for a specific server; let's use your finance department as an example. You can certainly collect resource statistics over a period of time in the middle of the month, and that might work just fine; however, the people in your finance department are more likely to utilize the system heavily during the first few days of the month as they are working on their month end processes. If you collect resource statistics on the 15th, you might miss a huge increase in resource utilization requirements, which could lead to the system not working as expected, making unhappy users.

One last thing before we jump into some example statistics; you should consider collecting these statistics over at least two periods for each server:

- First, during the normal business hours of your organization or the specific department, during a time when systems are likely to be heavily utilized
- The second round should include an entire day or week so you are aware of the impact of after hours tasks such as backups and anti-virus scans on your environment

It's important to have a strong understanding of the use cases for all the systems you will be virtualizing. If you are running your test during the middle of the month, you might miss the increase of traffic for systems utilized heavily only at the end of the month, for example, accounting systems. The more information you collect, the better prepared you will be to determine your resource utilization requirements.

There are quite a few commercial tools available to help determine the specific resource requirements for your environment. In fact, if you have an active project and/or budget, check with your server and storage vendor as they can most likely provide tools to assess your environment over a period of time to help you collect this information. If you work with a VMware Partner or the VMware **Professional Services Organization** (**PSO**), you could also work with them to run a tool called VMware Capacity Planner. This tool is only available to partners who have passed the corresponding partner exams.

For purposes of this section, however, we will look at the statistics we can capture natively within an operating system, for example, using Performance Monitor on Windows and the `sar` command in Linux.

If you are an OS X user, you might be wondering why we are not touching OS X. This is because while Apple allows virtualizing OS X 10.5 and later, it is only supported on the Apple hardware and is not likely an everyday use case. If your organization requires virtualizing OSX, ESXi 5.1 is supported on specific Mac Pro desktops with Intel Xeon 5600 series processors and 5.0 is supported on Xserve using Xeon 5500 series processors. The current Apple license agreement allows virtualizing OSX 10.5 and up; of course, you should check for the latest agreement to ensure you are adhering to the license agreement.

Monitoring common resource statistics

From a statistics perspective, there are four main types of resources you generally monitor: CPU, memory, disk, and network. Unless you have a very chatty application, network utilization is generally very low, but this doesn't mean we won't check on it; however, we probably won't dedicate as much time to it as we do for CPU, memory, and disk.

As we think about the CPU and memory, we generally look at utilization in terms of percentages. When we look at example servers, you will see that having an accurate inventory of the physical server is important so we can properly gauge the virtual CPU and memory requirements when we virtualize. If a physical server has dual quad core CPUs and 16 GB of memory, it does not necessarily mean we want to provide the same amount of virtual resources (you will learn why in the next chapter).

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Disk performance is where many people spend the least amount of time, and those people generally have the most headaches after they have virtualized. Disk performance is probably the most critical aspect to think about when you are planning your virtualization project. Most people only think of storage in terms of storage capacity, generally gigabytes (GB) or terabytes (TB). However, from a server perspective, we are mostly concerned with the amount of input and output per second, otherwise known as **IOPS** and throughput. We break down IOPS in into reads and writes per second and then their ratio by comparing one with the other. Understanding your I/O patterns will help you design your storage architecture to properly support all your applications. Storage design and understanding is an art and science by itself.

The *vSphere High Performance Cookbook*, *Prasenjit Sarkar*, *Packt Publishing* has an entire chapter dedicated to storage; *Troubleshooting vSphere Storage*, *Mike Preston*, *Packt Publishing* provides you with the key steps to help spot problems with storage from a vSphere perspective. Finally, one of the most in-depth books on storage is *Storage Implementation in vSphere 5.0 from VMware Press*, *Mostafa Khalil* (one of the original VMware Certified Design Experts in VCDX).

Sample workload

Let's break this down into a practical example so we can see how we are applying these concepts. In this example, we will look at two different types of servers that are likely to have various resource requirements: Windows Active Directory Domain Controller and a CentOS Apache web server. In this scenario, let's assume that each of these server operating systems and applications are running on dedicated hardware, that is, they are not yet virtual machines.

The first step you should take, if you do not have this already, is to document the physical systems, their components, and other relevant information such as computer or DNS name, IP address (es), location, and so on. For larger environments, you may also want to document installed software, user groups or departments, and so on.

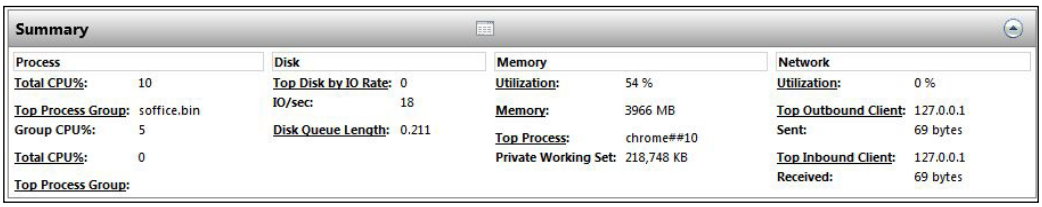
Collecting statistics on Windows

On Windows servers, your first step would be to start performance monitoring. Perform the following steps to do so:

1. Navigate to **Start** | **Run** and enter `perfmon`.

2. Once the **Performance Monitor** window opens, expand **Monitoring Tools** and click on **Performance Monitor**. Here, you could start adding various counters; however, as of Windows 2008/Windows 7, Performance Monitor includes Data Collector Sets.
3. Expand the `Data Collector Sets` folder and then the `System` folder; right-click on **System Performance** and select **Start**. Performance Monitor will start to collect key statistics about your system and its resource utilization.
4. When you are satisfied that you have collected an appropriate amount of data, click on **System Performance** and select **Stop**. Your reports will be saved into the `Reports` folder; navigate to **Reports | System**, click on the `System Performance` folder, and finally double-click on the report to see the report.

In the following screenshot for our domain controller, you can see we were using 10 percent of the total CPU resources available, 54 percent of the memory, a low 18 IOPS, and 0 percent of the available network resources (this is not really uncommon; I have had busy application servers that barely break 2 percent).



Summary							
Process		**Disk**		**Memory**		**Network**	
Total CPU%:	10	Top Disk by IO Rate:	0	Utilization:	54 %	Utilization:	0 %
Top Process Group:	soffice.bin	IO/sec:	18	Memory:	3966 MB	Top Outbound Client:	127.0.0.1
Group CPU%:	5	Disk Queue Length:	0.211	Top Process:	chrome##10	Sent:	69 bytes
Total CPU%:	0			Private Working Set:	218,748 KB	Top Inbound Client:	127.0.0.1
Top Process Group:						Received:	69 bytes

Now let's compare what we are utilizing with the actual physical resources available to the server. This server has two dual core processors (four total cores) running at 2 GHz per core (8 GHz total available), 4 GB of memory, two 200 GB SAS drives configured in a RAID1, and a 1 Gbps network card.

Here, performance monitor shows averages, but you should also investigate peak usage. If you scroll down in the report, you will find a menu labeled CPU. Navigate to **CPU | Process**. Here you will see quite a bit of data, more than the space we have to review in this book; however, if you scroll down, you will see a section called **Processor User Time by CPU**. Here, your mean (that is, average) column should match fairly closely to the report overview provided for the total, but we also want to look at any spikes we may have encountered. As you can see, this CPU had one core that received a maximum of 35 percent utilization, slightly more than the average suggested.

If we take the average CPU utilization at 10 percent of the total CPU, it means we will theoretically require only 800 MHz of CPU power, something a single physical core could easily support. The, memory is also using only half of what is available, so we can most likely reduce the amount of memory to 3 GB and still have room for various changes in operating conditions we might have not encountered during our collection window. Finally, having only 18 IOPS used means that we have plenty of performance left in the drives; even a SATA 7200 RPM drive can provide around 80 IOPS.

Collecting statistics on Linux

Now let's look at the Linux web server to see how we can collect this same set of information using `sar` in an additional package with `sysstat` that can monitor resource utilization over time. This is similar to what you might get from `top` or `iotop`. The `sysstat` package can easily be added to your system by running `yum install sysstat`, as it is a part of the base repository (yum install sysstat is command format).

Once the `sysstat` package is installed, it will start collecting information about resource utilization every 10 minutes and keep this information for a period of seven days. To see the information, you just need to run the `sar` command; there are different options to display different sets of information, which we will look at next.

Here, we can see that our system is idle right now by viewing the **%idle** column.

```
Linux 2.6.32-358.el6.x86_64 (li-ansible.lifiemage.lan)  11/22/2013      _x86_64_        (1 CPU)

10:50:01 AM     CPU     %user     %nice   %system   %iowait    %steal     %idle
11:00:01 AM     all      0.39      0.00      0.10      0.10      0.00     99.41
11:10:01 AM     all      0.38      0.00      0.11      0.12      0.00     99.39
11:20:01 AM     all      0.38      0.00      0.11      0.10      0.00     99.40
Average:        all      0.38      0.00      0.11      0.11      0.00     99.40
```

A simple way to generate some load on your system is to run `dd if=/dev/zero of=/dev/null`, which will spike your CPU load to 100 percent, so, don't do this on production systems! Let's look at the output with some CPU load. In this example, you can see that the CPU was under load for about half of the 10-minute collection window.

```
Linux 2.6.32-358.el6.x86_64 (li-ansible.lifiemage.lan)  11/22/2013      _x86_64_        (1 CPU)

10:50:01 AM     CPU     %user     %nice   %system   %iowait    %steal     %idle
11:00:01 AM     all      0.39      0.00      0.10      0.10      0.00     99.41
11:10:01 AM     all      0.38      0.00      0.11      0.12      0.00     99.39
11:20:01 AM     all      0.38      0.00      0.11      0.10      0.00     99.40
11:30:01 AM     all     11.13      0.00     42.87      0.05      0.00     45.95
Average:        all      3.07      0.00     10.80      0.09      0.00     86.03
```

One problem here is that unless the CPU spike, in this case to 100 percent, was not consistent for at least 10 minutes, we would potentially miss these spikes using `sar` with a 10-minute window. This is easily changed by editing `/etc/cron.d/sysstat`, which tells the system to run this every 10 minutes. During a collection window, one or two minutes may provide more valuable detail. In this example, you can see I am now logging at a five-minute interval instead of 10, so I will have a better chance to find maximum CPU usage during my monitoring period.

```
Linux 2.6.32-358.el6.x86_64 (li-ansible.lifiemage.lan)  11/22/2013      _x86_64_        (1 CPU)

10:50:01 AM     CPU     %user     %nice   %system   %iowait    %steal     %idle
11:00:01 AM     all      0.39      0.00      0.10      0.10      0.00     99.41
11:10:01 AM     all      0.38      0.00      0.11      0.12      0.00     99.39
11:20:01 AM     all      0.38      0.00      0.11      0.10      0.00     99.40
11:30:01 AM     all     11.13      0.00     42.87      0.05      0.00     45.95
11:40:01 AM     all     11.99      0.00     46.73      0.03      0.00     41.25
11:45:01 AM     all      0.42      0.00      0.21      0.09      0.00     99.28
Average:        all      4.46      0.00     16.38      0.08      0.00     79.08
```

Now, we are not only concerned with the CPU, but we also want to see memory and disk utilization. To access those statistics, run `sar` with the following options:

- The `sar -r` command will show RAM (memory) statistics. At a basic level, the items we are concerned with here would be the percentage of memory used, which we could use to determine how much memory is actually being utilized.
- The `sar -b` command will show disk I/O. From a disk perspective, `sar -b` will tell us the total number of transactions per second (tps), read transactions per second (rtps), and write transactions per second (wtps).

As you can see, you are able to natively collect quite a bit of relevant data about resource utilization on our systems. However, without the help of a vendor or VMware PSO who has access to VMware Capacity Planner, another commercial tool, or a good automation system, this can become difficult to do on a large scale (hundreds or thousands of servers), but certainly not impossible.

Summary

In this chapter, we looked at the various resource requirements to build a working vSphere environment, including the resource requirements for both ESXi and vCenter. We also reviewed the *VMware Compatibility Guide* to validate the hardware where installing ESXi would be supported.

We then wrapped up by looking at the various resources we want to support when we create virtual machines to ensure the performance is acceptable and within your organization's SLAs. What we saw in the two examples is not different from what many organizations face; they have invested in resources by purchasing these servers. However, it will not receive the full **Return on Investment** (**ROI**) because the systems are not being more heavily leveraged.

In the next chapter, we will see how VMware vSphere can utilize server resources to provide your organization with a great ROI efficiently while most likely decreasing its TCO because of its ability to consolidate several physical machines onto a single host. This provides greater availability that you could achieve without the use of VMware vSphere.

2
Assigning Resources to VMs

Now that you understand the various resource requirements for installing ESXi and vCenter, we will look at how VMware vSphere manages resources assigned to an individual VM and how multiple VMs on a single physical host can affect resource availability. Throughout this chapter, we will compare and contrast common practices used while deploying operating systems to bare metal systems and then in deploying operating systems as VMs.

The topics we'll be covering in this chapter are as follows:

- The basics of overcommitment and virtualization
- CPU scheduling and the effect of multiple vCPU VMs
- Memory assignment and management
- Storage considerations and their effects on performance
- Networking, Virtual Switches, VM to VM, and VM to physical system communication

The basics of overcommitment and virtualization

As hardware evolved, processors and memory far outpaced the rate that applications could consume those resources. When multicore processors came out, many applications were still only able to utilize a single core, that left an entire CPU core idle unless some other application or process was multiprocessor capable. While an underutilized resource provides headroom for the applications, it reduces value to the business because the resource does not receive full returns on its hardware investment. When a system or application was idle, the investment made by the organization was further wasted.

Virtualization helps to solve this problem by being able to assign physical resources to virtual servers. VMware made this technology available to all organizations. It simplified deployment and built features that could be leveraged by a small organization with only a handful of servers, or by the world's largest organizations with thousands or even tens of thousands of servers.

Using the two example servers from *Chapter 1*, *Understanding vSphere System Requirements*, let's consider the benefits of virtualizing your servers compared to installing them on dedicated hardware.

We looked at two physical servers, each with a dual core CPU, so two processing cores. The Windows server, even at its peak utilization, never used more than 35 percent of the available processing power of the server, leaving 65 percent of the resources idle. Our Linux server (also with a dual core CPU) was almost entirely idle, and even when we added a workload, we still only averaged out 45 percent CPU utilization. This means your organization purchased two physical servers, while the application only required the resources available in one physical server.

Let's take this example even a step further. While the Windows server required a peak of 35 percent of the CPU, it did not require this for the entire duration in which we collected the resource utilization statistics. If we assume for the sake of this example that the spike on the Windows server and on the Linux server happened at separate times, we would have even more resources available if we had other VMs running on the same physical hardware.

The same concept of sharing physical resources carry through the four major resources that we discussed in *Chapter 1*, *Understanding vSphere System Requirements*, that is, CPU, memory, disk, and networking. It's unlikely that a single OS and application are utilizing all of the physical resources available in a modern server. By virtualizing several OS and applications, we can maximize the resource utilization available in a physical server, providing a higher ROI and in many cases, a lower TCO for the organization.

CPU scheduling and the effect of multiple vCPU VMs

Modern CPUs are so powerful that we perceive the ability to multitask or use several programs at once. In reality, CPU cores act in a very serial fashion, that is to say, a single core is only ever really executing one task at a time. Because of the speed and logic in these modern processors, they are able to quickly switch between the requests from different applications, so when one request is finished, the CPU switches to the next task and so on.

A technology such as **Hyper-Threading** (**HT**) by Intel or AMD-V/RVI will make a single core appear as two cores to the operating system and can help process multiple requests at once on a single CPU core.

If you have HT enabled, you will see that your OS, if supported with a hypervisor such as ESXi, perceives this to be an available physical processor core; however, for purposes of designing your infrastructure, I would recommend staying within the physical capabilities of your processor. With an Intel Xeon CPU running HT on four cores, the ESXi hypervisor will be presented with eight available cores. Although HT increases number of the available virtual cores, you should design your clusters for the load based on the number of physical cores.

With this basic understanding of how CPUs function, you can see why it's important to understand the resource requirements for a specific server or application. Servers or applications with relatively low CPU resource requirements could potentially work fine with 10 or 12 vCPUs per physical core; however, if the application was processor intensive, you might find that you can only run two to three vCPUs per physical core. It's important to note here that generally, you are not assigning a VM or vCPU to a physical CPU; however, you could through the use of CPU affinity in ESXi. We are simply saying that depending on your workload and the number of physical cores in your CPU, you could expect to receive certain consolidation ratios.

When possible, you should try to perform the following actions:

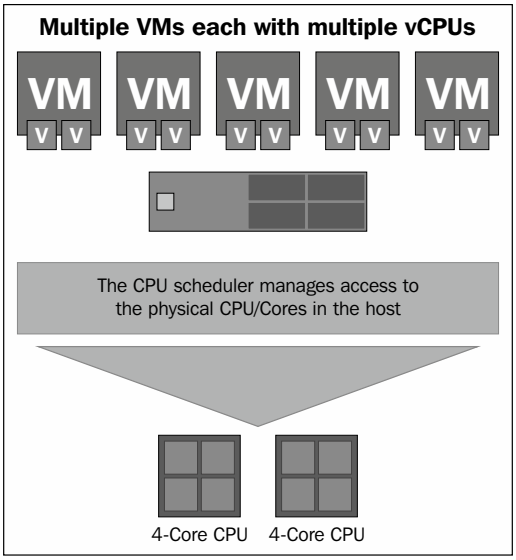
- Monitor resource utilization prior to any project—physical to virtual or virtual to virtual
- Do not overcommit the CPU; only assign what the VM and workload require
- Avoid CPU affinity as you will not be able to vMotion VMs with CPU affinity enabled until it is disabled

VMware has developed an advanced CPU scheduler to ensure the efficient operation of VMs in your environment. In fact, they have published an excellent technical white paper that you should read, as we don't have the space in this book to cover the CPU scheduler in that level of detail. You can find the white paper at `http://www.vmware.com/files/pdf/techpaper/VMware-vSphere-CPU-Sched-Perf.pdf`.

If your VM has one vCPU, and the host it is running on has four pCPU, then it's the job of the CPU scheduler to find an available pCPU for the VM to access when required. That is an easy translation to make to a physical server with an OS installed. However, what happens when your VM has multiple vCPUs?

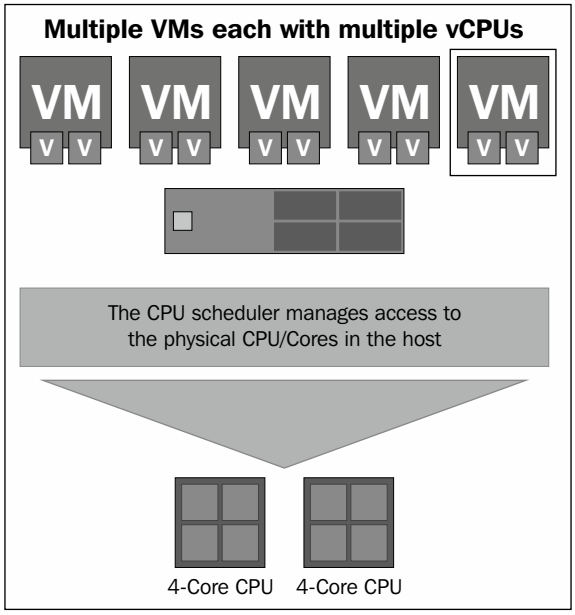
Using vCenter as an example, as we discussed in *Chapter 1*, *Understanding vSphere System Requirements*, vCenter recommends at least two vCPUs. However, at any given point, it may not require the processing power of both vCPUs. If vCenter was installed on its own physical server, the OS would expect to have access to all of the CPUs or cores on that server and would just operate as required. However, in a VM where you are sharing a pCPU between multiple VMs, the pCPU might already be busy processing a request from another VM. Since the OS expects all of the CPUs assigned to it to be available, it must have to wait for two pCPUs to be available since it is assigned two vCPUs. Let's look at another example visually.

In the following figure, we have an ESXi host with two quad core pCPU for a total of eight pCPU cores available to VMs with five VMs, each with two vCPUs assigned:



When all eight cores are available and the first VM makes a request, the CPU scheduler can provide the vCPUs with access to two pCPU cores, so the VM's operating system can operate as expected. The CPU scheduler can support simultaneous requests for only three more VMs running on this host. When the fifth VM makes a request, the CPU scheduler does not have access to any pCPU cores. Because the VM is unaware that it is virtual, it assumes it has access to the number of processors assigned.

In this scenario, which is shown in the following figure, the VM will have to wait until all of the assigned vCPUs have access to pCPUs to service the request. Even if only one vCPU is required, the OS has to wait until all vCPUs can be serviced.



This is where the concept of rightsizing comes into play. Prior to virtualization, the OS had access to all pCPUs in a server, whether or not it needed this access. So, a web server with dual quad core processors could generally be idle a majority of the time depending, of course, on its workload. If we used the same configuration in a virtual machine, that is, assigning eight vCPUs, the VM would have to wait until the CPU scheduler could provide access to eight pCPUs at once, whether or not it needed all eight pCPUs.

Going back again to the example at the end of the previous chapter, where we had a physical server with two cores utilizing only 35 percent of the CPU, you would be better off assigning only one vCPU to the VM for the same workload. This would help since the scheduler would only ever need to access one pCPU that could provide all the necessary computing power that the workload required.

To build off yet another topic that we covered in *Chapter 1*, *Understanding vSphere System Requirements*, it's important to understand the resource requirements for environment and specific applications. Once virtual, you want to make sure you provide the required resources to a VM and do not carry over the practice from the physical world of providing extra resources that are not required. There are quite a few products in the market today to help you identify VMs that are over-provisioned, some are even free of charge.

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One item that you should be aware of once virtualized is a command called `ESXTOP`. `ESXTOP` is a command-line utility to monitor resource utilization for your ESXi host and the VMs on that host. If you are having performance problems with a VM and you want to see whether it's due to the pCPU resources that have not been made available to the VM, you can check **%RDY**.

%RDY displays the time span that your VM is ready to run for, but the scheduler was not able to provide pCPU resources. This should be under five percent in normal operation. A number higher than five percent could mean you have overcommitted your pCPU or assigned too many vCPUs to a VM.

If your VM has multiple vCPUs assigned, you should also monitor **%CSTP**, which is only relevant for multiple vCPU VMs. A VM with a %CSTP of 3 percent or higher suggests you should lower the number of vCPUs assigned to the VM. In the case of a VM that has four vCPUs assigned but operates at 30-40 percent CPU utilization, you might actually improve performance by removing two vCPUs and leaving the VM with two vCPUs. Here, the VM operates at 40 percent capacity,
so removing 50 percent of the CPUs should leave enough compute power for the VM to operate.

Other useful statistics include (host or VM-based statistics) the following:

- **CPU load average**: This gives the average utilization of all pCPUs in the host. A reading of 1.0 in the host means you are fully utilized, 0.5 means 50 percent utilized, 0.02 means two percent utilized, and so on.
- **PCPU USED (%)**: This is the percent of all CPUs used by the host.
- **%RUN**: This is the amount of vCPUs utilized by the VM. This setting takes into account all vCPUs that are assigned. So, a value of 100 percent with one vCPU means that the vCPU is fully utilized. A value of 150 percent for a VM with two vCPUs means that it is roughly 75 percent utilized if you are looking within the guest OS (VM).
- **%WAIT**: This refers to how long the VM is waiting for other processes managed by ESXi to complete, such as I/O (VM).

To dive deeper into ESXTOP how-tos, check out VMware KB 2001003 at `http://kb.vmware.com/kb/2001003` and the VMware communities post at `https://communities.vmware.com/docs/DOC-9279`.

Memory assignment and management

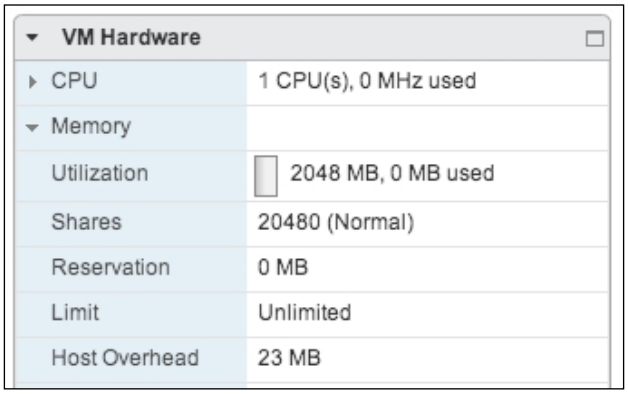
While physical memory is allocated to VMs through the hypervisor just like CPU, the calculations and methods to determine memory requirements are much simpler. Memory like vCPUs, are assigned to a VM, and the OS on the VM assumes it has that amount of memory available to it; however, in many scenarios, you are likely to overcommit the memory to achieve a greater consolidation ratio or increase the VM density per host. The benefits of overcommitment for memory are again along the same lines as overcommitting CPUs; your VMs are not likely to use the entire amount of memory assigned to them at all times. As with all resources in a virtual environment, you want to rightsize your memory configuration as well. While you could easily give all VMs 16 GB of memory, the amount of memory assigned to a VM has other impacts on your infrastructure and the vSphere cluster, which we will see shortly. VMware has several memory management techniques to reduce the likelihood of contention or in the event that you run into a period of contention techniques to reclaim memory from VMs.

A current trend, thanks to decreasing memory costs and the manufacturer's ability to increase the amount of memory available within a system, is actually to not overcommit on memory. Take for example, a modern yet mid-range physical server such as a Dell R520 or Cisco UCS B22M blade that can contain up to 384 GB of memory. If you were to configure the VMs on these servers each with 4 GB of memory, you would have enough memory capacity for around 92 to 94 VMs, leaving enough memory for ESXi itself. Depending on the CPU you selected, anywhere from four to eight cores per processor, you would potentially be pushing a 12:1 vCPU to pCPU overcommitment ratio if each VM had a single vCPU. In this scenario, you might actually find CPU cores to be your limiting resource instead of memory. In this example, however, we have not factored in the memory required to run ESXi or the memory overhead for the VM. What is memory overhead you ask? Well, we were just getting to that.

Memory overhead

Memory overhead is additional memory that each VM requires in order for the ESXi to manage it. It is generally a very small amount of memory based on the amount of memory assigned to the VM and the number of vCPUs. Using the vCenter requirements from *Chapter 1*, *Understanding vSphere System Requirements*, a VM with two vCPUs and 16 GB of memory would require about 144 MB of memory overhead. Another example, a VM with eight vCPUs and 16 GB of memory should only require around 169 MB of memory overhead.

You can see the memory overhead for any particular VM on the summary page by expanding **Memory**, as seen in the following screenshot from the vSphere 5.5 web client:



You can also find a sample memory overhead charge in the *VMware Resource Management* guide at `http://pubs.vmware.com/vsphere-55/topic/com.vmware.ICbase/PDF/vsphere-esxi-vcenter-server-55-resource-management-guide.pdf`. The following chart from VMware is meant to serve as a guide to help you determine memory overhead based on the resources assigned to the VM. It is useful while designing your infrastructure, but there are other advanced settings that also factor into memory reservations, such as the VMX swap file.

Memory (MB)	1 vCPU	2 vCPUs	4 vCPUs	8 vCPUs
256	20.29	24.28	32.23	48.16
1024	25.90	29.91	37.86	53.82
4096	48.64	52.72	60.67	76.78
16384	139.62	143.98	151.93	168.60

Transparent page sharing and memory compression

Two memory management techniques that ESXi implements to improve memory utilization is **transparent page sharing** (**TPS**) and memory compression. As you might expect from its name, ESXi compresses memory pages during times of contention to prevent swapping. This is done because the process to compress and thus decompress the memory when required is still faster than swapping to a disk. Memory compression is enabled by default, and unless you have been instructed to do so by support, it's a setting best left enabled.

Transparent page sharing is where identical memory pages are shared among multiple virtual machines. There is some debate as to the usefulness of TPS as more guess, operating systems are leveraging large memory pages, typically 2 MB in size compared to 4 KB.

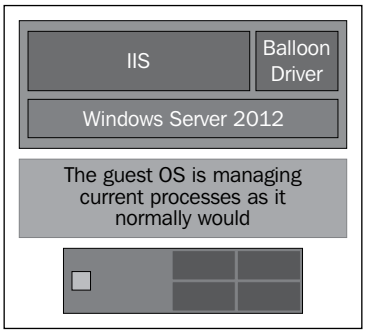
With 4 KB page sizes, it is theoretically easier to find a match among multiple memory pages, which ESXi can then basically deduplicate. If ESXi were to find 1000 4 KB pages, it could reduce that to a single memory page, thus making more memory available to VMs and other processes. With large pages, ESXi still scans the page but rather than trying to match an entire 2 MB page size, which is less likely, it scans 4 KB pages within the 2 MB page. If ESXi detects contention, it can then attempt to share, compress, or swap these smaller pages. For the purposes of this book, you should know that TPS attempts to deduplicate and share identical memory pages. For a deeper dive into TPS, VMware has published a white paper devoted entirely to TPS, which can be found at `http://download3.vmware.com/software/vmw-tools/papers/WP-2013-01E-FINAL.pdf`. You should also read *VMware vSphere 5.1 Clustering Deepdive*, *Duncan Epping* and *Frank Denneman*. This is hands down one of the best available technical books on VMware vSphere. I'll be suggesting this book again in the next chapter when we discuss HA and DRS.

Unless required by your application, leaving large memory pages disabled allows TPS to more efficiently deduplicate memory.

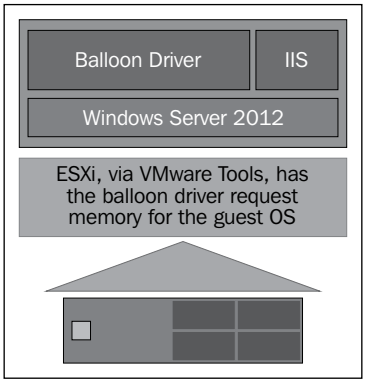
Ballooning

There is a driver installed on your guest OS when **VMware Tools** is installed. This allows ESXi to request memory from VMs that might otherwise be idle. ESXi asks the balloon driver, a process that is running in the guest OS, to request memory from the OS as any other application might. This allows the guest OS, which is aware of its current workload, to allocate memory without negatively impacting the performance of active processes.

Take, for example, a Windows server running IIS as shown in the following figure; an increase in demand may cause IIS and its related processes to require a specific amount of memory.



The memory that IIS previously requested may not be returned to the OS immediately even though it is no longer required. If ESXi is experiencing memory contention, ESXi makes a request to the balloon driver to ask the guest OS for additional memory. The guest OS can then determine what, if any, memory can be reclaimed from other processes. The OS can reclaim the memory not needed from other applications and provide it to the balloon driver, which then notifies ESXi of how much memory can be reclaimed from the guest. By default, ESXi is configured to reclaim up to 65 percent of the assigned memory, and this can be alerted with advanced settings at both the ESXi hosts, thus affecting all VMs or individual VMs. Make sure VMware Tools is installed on all VMs to allow ballooning to function.



The VSWP swap file

The VSWP swap file is a file equal to the amount of memory assigned to the VM minus the amount of memory reserved (we will discuss reservations in *Chapter 3*, *Advanced Resource Management Features*, but it's very literal). The swap file is one of the methods used to reclaim memory during times of contention. When this happens, ESXi swaps the running memory to disk, much like what you would see with the Windows swap file or Linux swap partition. It should be noted that OS swapping is separate from the VSWP swap file, and your OS will still utilize its native swapping techniques if the guest OS is low on memory. ESXi will utilize the VM swap file if it is low on memory, even if the VM itself is not low on memory.

The VSWP swap file is stored by default in the same location as your virtual hard drive. When you think about this, as it relates to rightsizing, you can see why assigning too much memory for no reason can have an impact on your environment. If you had 100 VMs running on your host, each with 16 GB of memory and no memory reservations, you would need an additional 1.6 TB of storage space to store all of your VSWP swap files. If those VMs really needed 4 GB of memory, then you would only need an additional 400 GB of storage, which is not a trivial amount but much less than the 1.6 TB.

> Like CPU, it is important to rightsize the VM memory configuration.

Do not forget to plan for memory overhead as well as additional storage requirements for the VSWP swap file.

While swapping is generally considered a bad thing, after all, it means we are running low on memory in our ESXi host; the memory management techniques used in advanced can typically eliminate the need for swapping if planned properly. If possible, place the VSWP swap file on a separate storage, preferably flash, for the best possible performance during memory contention.

Monitoring memory usage

As we discussed in case of the CPU, there are several useful `ESXTOP` statistics for monitoring memory. When you run `ESXTOP`, you are given the CPU view, which is the default view. To switch to memory statistics, just hit *M* on your keyboard and the view will change. Refer to the VMware communities list again and the KB article that we mentioned earlier if you need a refresher (`https://communities.vmware.com/docs/DOC-9279` and `http://kb.vmware.com/selfservice/microsites/search.do?language=en_US&cmd=displayKC&externalId=2001003`). The first item I tend to look at on a host is **MEM overcommit avg**. It shows how much memory you have overcommitted on your host.

A host that is not overcommitted (which is when the host has more memory than the VMs running on the host are assigned plus memory overhead) will have a return value of 0 as shown in the following screenshot:

```
 4:38:33pm up 29 days  3:13, 497 worlds, 4 VMs, 7 vCPUs; MEM overcommit avg: 0.00, 0.00, 0.00
PMEM  /MB: 131062   total:  1877     vmk, 18815 other, 110369 free
VMKMEM/MB: 130676 managed:  1921 minfree,  5888 rsvd, 124788 ursvd,  high state
NUMA  /MB: 65536 (59741), 65524 (50244)
PSHARE/MB:    22  shared,    21  common:     1 saving
SWAP  /MB:     0    curr,     0 rclmtgt:                 0.00 r/s,   0.00 w/s
ZIP   /MB:     0  zipped,     0   saved
MEMCTL/MB:     0    curr,     0  target, 16201 max
```

You have probably noticed there are three values of **0.00**, **0.00**, and **0.00** returned in the preceding screenshot. These are the averages over a 1, 5, and 15 minute window. The other statistics that you should make note of are as follows:

- **PSHARE**: This is a count of the number of shared pages and memory saving due to TPS.
- **SWAP**: This is the total amount of memory being swapped to the disk. This will most likely cause noticeable performance problems and should relate to a high **MEM overcommit avg**. Consider adding more memory or reducing the number of VMs on the host if this is consistently high.
- **MEMCTL**: This shows how much memory has been reclaimed by the balloon driver. A consistently high value here could lead to future disk swapping; however, this is a valid memory overcommitment management feature. You should also consider monitoring your individual VM OS for swapping if this value is high, to ensure VM performance is optimal.

There are also statistics to monitor at the VM level. For example, if **MEMCTL** is high, you may want to add the **MCTLSZ** column to see which VMs are returning memory via the balloon driver. To add this column that is not visible by default, perform the following actions:

1. Hit the *F* key on your keyboard.
2. Hit the *J* key on your keyboard to select **MCTL**.
3. Hit *Esc* and you will be returned to the `ESXTOP` statistics view with the new columns.

Now you can see how much memory has been reclaimed (if any) or could be reclaimed by the balloon driver by examining the **MCTLSZ** and **MCTLMAX** columns.

Storage considerations and their effects on performance

Storage is the one area that you cannot skip while evaluating your environment and workload, and no amount of CPU or RAM will make up for a misconfigured or underconfigured storage system. Storage in many environments is the biggest resource bottleneck due to its cost, general misunderstanding about how it works, and what is important. Outside of IT, storage is typically thought of in terms of capacity which, while a valid point, is only a small piece of the storage equation.

From a business perspective, cost is the driving factor on hard drives. Our job as administrators, engineers, and architects is to understand the workload so that we can select the storage appliances, drives, and connectivity that will support the current environment as well as future growth. You could spend an entire career learning about and mastering storage, but we have only few pages in this book. For a deeper dive into storage, I would suggest picking up *Storage Implementation in vSphere*, *Mostafa Khalil* and *Troubleshooting vSphere Storage*, *Mike Preston*.

What is IOPS?

For the purposes of our discussion, let's start at the storage appliance. In most vSphere environments, storage is provided by hardware and networks separate from the physical server known as the storage fabric. Two of the main metrics that we will look at is **I/O per Second** (**IOPS**) and throughput (the amount of time it takes for data to be transmitted). Let's start with IOPS, which is mostly determined by the hard drive, which also makes it one of the most important factors when it comes to performance. Typically, you will see four types of hard drives that are available (from the slowest and largest to the fastest and smallest); for example, SATA (or NL-SAS), SAS, FC, and SSD.

While SATA drives can generally provide enough storage capacity, they cannot generally provide enough IOPS, and SSDs can provide enough IOPS but generally not enough capacity. The following is a chart of generally accepted average hard drive IOPS:

Drive Type	IOPS	Capacity
SATA 7200 RPM	Up to 100 IOPS	Up to 4 TB
SATA 10,000 RPM	Up to 150 IOPS	Up to 1 TB
SAS 10,000 RPM	Up to 140 IOPS	Up to 1 TB
SAS 15,000 RPM	Up to 210 IOPS	Up to 600 GB
SSD	Up to 400 to 120,000 IOPS	Up to 1.2 TB

RAID

Unlike in a laptop or desktop where you usually have a single hard drive, storage appliances are composed of multiple drives grouped together called arrays and more specifically into a **redundant array of independent disks** (**RAID**), though you may also see **Just of Bunch of Disks** (**JBOD**), which does not provide any redundancy.

The IOPS of your array is the number of drives of that array multiplied by the IOPS of the drive minus any RAID write penalties.

You should check with your storage appliance manufacturer to determine the recommended number of drives to place in a single array, which may vary based on the type of array you created.

Once the RAID or JBOD is created, you can then create a **LUN**. Depending on your storage appliance, you could create a single LUN on that RAID, or you could create multiple LUNS. Whenever possible, I prefer to create a single LUN on a single RAID or JBOD that fulfills the workload requirements being run by the VMs. By doing this, you can easily identify the workload and the associated VMs running that workload for a particular LUN. If you were to create multiple LUNS on a single RAID or JBOD, you may run into contention when a workload on one LUN is utilizing all of the I/O resources available from that RAID, which may not be immediately obvious while observing perceived performance problems on the other LUN.

RAID has a major impact on your available IOPS and adds another layer of complexity to account for workload patterns. Depending on your application, you could have read a heavy workload, that is, the one that is reading more data than it is writing, such as a reporting application, or a write heavy workload, that is, the one that performs more writes, such as applications that import large amounts of data.

You may even have applications where your workload changes depending on business cycles. One such example is a financial system which may have a write-heavy workload at the end of a month when the information is being imported and which then changes to a read-heavy workload at the beginning of the month when the reports are being run.

While we don't have enough space to dive deep into RAID types, the following chart should give you what you need to begin your research. Depending on your storage appliance, you may be able to assign what is called a hot spare drive, which will be used to automatically replace a failed drive in a RAID, though generally it's only useful where there is already a redundancy.

RAID Type	Redundancy	Read or Write Favorable
RAID 0	No	Both
RAID 1	Yes; 1 disk	Both, limited by the disk type
RAID 5	Yes; 1 disk	Read
RAID 6	Yes; 2 disks	Read
RAID 0+1	Yes; depends on RAID 0	Both

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RAID 5 is one of the more common RAID types deployed because of its ability to maximize storage capacity and provide failure of a single disk without losing data. In nonvirtualized environments, RAID 5 could typically provide enough IOPS for even a write-heavy workload since it was the only workload utilizing those drives. The downfall of RAID 5 (and RAID 6) is that in order to achieve the drive redundancy, extra data is written that creates what is known as a write penalty since a single write has to be performed on two drives: the drive that the write was originally intended for and a parity write on a separate disk, which provides the redundancy. In a virtual environment where the LUN may be servicing multiple workload, the write penalty is magnified even further.

Selecting the drive type as well as the RAID type go hand in hand. Most modern storage appliances employ a data tiering system where you can perform writes on faster drives such as SAS or SSD, and the data is then moved to the most cost-effective SATA drives after a certain period of inactivity. Additionally, an increasing number of vendors are offering all-flash arrays that can provide extremely high levels of IOPS but with lower storage capacity; depending on your workload, this may be sufficient.

VMware vSphere Storage APIs – Array Integration (VAAI)

Most storage appliance vendors will partner VMware to offer what is called VMware vSphere Storage APIs – Array Integration or VAAI. VAAI allows VMware vSphere to offload certain tasks that would otherwise consume host resources such as cloning a VM to the storage appliance. These tasks, called primitives, can be managed more efficiently by the storage appliance by itself. When you are researching storage options for your vSphere environment, you should definitely consider appliances that support VAAI. You can find more information about VAAI at `http://www.vmware.com/files/pdf/techpaper/VMware-vSphere-Storage-API-Array-Integration.pdf`.

The following are some points to consider while determining your storage requirements:

- Spend extra time monitoring I/O and throughput requirements
- Select drives and RAID options that will fit your IOPS requirements today as well as allow for future expansion
- Understand your VM and application workload profile, that is, read or write-heavy and possibly both at different times
- Use storage appliances and vendors that support VAAI

Connectivity and throughput

Once you have your drive type and RAID type designed to support your workload, you then need to consider how you will connect your storage appliance(s) to your physical servers. There are three options for connecting your storage appliance: Ethernet, which is the same as your typical networking equipment, that supports iSCSI, NFS and **Fiber Channel over Ethernet** (**FCoE**); **Infiniband**; and traditional **Fiber Channel** (**FC**). Infiniband currently provides the fastest connectivity, which will improve throughput but it is also the most costly. Having said that, nothing costs more than outages and negative business impact. If your workload calls for Infiniband, you should advocate it. FC has been generally provided the best balance between cost and performance, but as 10 GbE (10 Gigabit Ethernet) becomes more cost effective, you might see a shift towards FCoE, iSCSI, and NFS.

Since you are typically connecting your storage appliance to multiple physical servers, a switch is used just as you would use a switch for networking purposes. Most designs implement a completely separate storage switch; however, many modern switches support a modular architecture where you can combine Ethernet for both networking and storage as well as FC or Infiniband into the same switch. What you choose for your environment should largely depend on your workload requirements; if you are looking to cut budget, you should probably look at areas other than your storage fabric.

Throughput, which will be affected by the type of connectivity you use, is the measure of how much data can be written. While IOPS measures the number of read/write operations per second, throughput measures how many KBs, MBs, or GBs can be written in a certain amount of time, typically measured in MB per second. There are many factors within your storage appliance which will also affect throughput, so check with your storage vendor and ask them to help you determine what your workload profile is, and test that profile against their storage array.

VMFS

With most connectivity selections, you will format your storage in the Virtual Machine File System format (VMFS); however, if you have gone down the NFS route, then NFS will be your filesystem format. There are a few items to be aware of with NFS. First, you will thin provision your virtual hard drives (VMDK); thin provisioning consumes only the amount of space actually required by the virtual hard disk. If you assign a 100 GB virtual hard drive to a VM but it only uses 20 GB, you will only consume 20 GB of actual storage. We will touch on the pros and cons of thin provisioning shortly. Also, depending on your NFS network and uplink configuration, you may not be taking advantage of multiple uplinks for better performance. This is because the default load balancing method uses both the

source and destination IP addresses, even with multiple bonded uplinks; these will always be the same if the NFS targets are all presented on the same IP address from your storage appliance. There are a few options to achieve better load balancing, such as creating multiple NFS servers with unique IP addresses. The VMware *Best Practices for Running VMware vSphere on Network-Attached Storage* document has more information on NFS considerations at (`http://www.vmware.com/files/pdf/techpaper/VMW-WP-vSPHR-NAS-USLET-101-WEB.pdf`).

While the items mentioned previously may seem like limitations, they are simply considerations that you will also have to plan for and manage with block-based protocols such as FC, FCoE, and iSCSI. With these protocols, you still need to design your storage fabric to provide the necessary throughput and redundancy. The debate between block and NFS, which for many years favored block, has come down to requirements and supportability. What your infrastructure requires and how well can you support the infrastructure should ultimately dictate which storage protocol you select.

VM disk provisioning

While NFS makes the decision on thin provisioning for you, this is still an option with VMFS, and it's important to understand the differences. As mentioned in the previous section, thin provisioning only allocates the amount of space actually used within the VMDK. This allows you to overcommit your storage appliance, like we have discussed with the CPU and memory. This can be useful where growth is expected over a long period of time or when the VM is not easily taken offline for maintenance. While the overhead of expanding a thin provisioned disk has been tested to be negligible, applications with high write workload may still be impacted. In these scenarios, you may wish to thick provision your VMDK.

Thick provisioning, as you may have guessed, allocates the full amount of assigned storage immediately. If you create a VMDK with 100 GB of space, that 100 GB of space is allocated on your storage appliance immediately. There are two options available for thick provisioned VMDKs: Eager Zeroed and Lazy Zeroed. Lazy Zeroed allocates the entire amount of space immediately to the VMDK but does not wipe or zero the blocks until the first write request. With Eager Zeroed disks, all blocks are wiped when the disk is created. This causes the disk creation time to take longer but provides the best possible performance for write-intensive applications. For Windows administrators, you can draw a parallel to quick formatting of an NTFS partition to thick Lazy Zeroed versus a full format to an Eager Zeroed drive.

Monitoring storage

Troubleshooting storage can be quite difficult; however, like CPU and memory, there are some key ESXTOP statistics to look at if you suspect a storage bottleneck. To access disk statistics, hit *v* on your keyboard while in ESXTOP. Unlike CPU and memory, this will be strictly VM-focused. The following counters are focused specifically on VMs:

- **READS/s**: This is the number of reads per second
- **WRITES/s**: This is the number of writes per second

The sum of these statistics should match the expected I/O from your datastore and LUN/array. For example, if you had seven SATA drives in a RAID 5, and your READS/s and WRITES/s were equal, that is a 50/50 split, then you could expect to get somewhere around 224 IOPS from that datastore/LUN/array (if you kept to a single datastore per array, otherwise you need to view all VMs on all datastores/LUNs created on that array). If the combination of reads and writes exceeds 224, then you have more I/O than the LUN can handle.

As we mentioned earlier, I/O is not the only factor in performance, you also need to consider throughput. To view storage adapter-related statistics, press the *d* key on your keyboard while in ESXTOP. The statistics to monitor include the following:

- **GAVG**: This is a measure of the round trip latency seen by the VM.
- **KAVG**: This is the latency related to the ESXi kernel.
- **DAVG**: This is the latency seen at the device driver level. It includes the roundtrip time between the HBA and the storage.

These statistics would likely need to be related to similar statistics from your storage array to determine whether the bottleneck is host-related, a specific array, a storage appliance, or a storage network.

Networking

Networking in a vSphere environment is not all that different from physical networking. VMs are assigned virtual network cards which are attached to switches. A virtual switch has an uplink to a physical switch.

Uplinks

An uplink is a physical NIC on the ESXi hosts that connects the server to the network. In a typical ESXi deployment, you typically have multiple physical NICs to support various types of network traffic. It is also considered best practice to have redundancy for your various uplinks with multiple NICs supporting a single traffic type. Depending on your requirements, this may be a single active NIC with a standby NIC, which becomes active in the event of a failure, or you may bond multiple interfaces together and create an LACP group on your physical switch to support additional throughput. In extreme cases, I have even heard of architects using NICs from different manufacturers so that a driver failure or crash would not affect all NICs on a host.

Having multiple physical NICs/uplinks is only part of the equation. You also need to determine where these uplinks will connect to. If you have used multiple physical nicks for redundancy, but only one uplink to a single physical switch, then you still have a single point of failure in the network stack.

Wherever possible, consider the following requirements:

- A minimum of two physical NIC ports, preferably on two physical NICs for each type of traffic uplink (for example, one on board the NIC port and one on the PCI card port)
- Each pair of NIC ports should connect to separate physical switches or at least separate line cards in a modular switch
- If bonding multiple NIC ports together for throughput, verify the recommended configuration from your switch vendor

At a minimum, there are two types of traffic from an ESXi host: VM kernel and VM network traffic. VM network traffic is just that—the network traffic from your VMs to the physical network. VM kernel or management traffic is the interface used to manage the ESXi host, connect it to vCenter, and for advanced features such as vMotion or Fault Tolerance; we will cover vMotion and Fault Tolerance in the next section.

As we have said before, the network design ultimately depends on your specific requirements. You could certainly run your management and VM network traffic over the same uplink, or you could just easily have separate physical uplinks for management, vMotion, FT, and VM network and storage; beyond that, you may even have multiple vSwitches for spreading out the same type of traffic.

What is a vSwitch?

A vSwitch is quite similar to a physical switch. A vSwitch has ports that VMs connect to as well as various settings that you might also configure on a physical switch, such as VLANs, or port groups on a vSwitch, and MTU settings if you need to configure jumbo frames. There are two types of vSwitches: a Standard Switch (vSS) and Distributed Switch (vDS)—think of a vSS like a typical 1U 24 or 48 port switch that has some management features. A vDS is more like an advanced enterprise grade switch that can support multiple line cards with more advanced features. Having said that, there is no reason why a vSS cannot be used in large environments. As we have said many times, it comes down to your requirements and whether the features you would like to use are available on a vSS or on a vDS.

Like physical switches, vSwitches support VLAN tagging to allow you to separate different types of network traffic. On a vSwitch, you would create a port group to support different VLANs. If you are using a vSS, you need to create vSwitches and port groups on each host. With a vDS, all port groups are managed centrally via vCenter and pushed to each ESXi host. In the case of a vDS, each ESXi host acts in a similar fashion to line cards in a modular switch. While vCenter is critical for managing a vDS, it will continue to operate if vCenter is unavailable, but you will not be able to make any changes until vCenter is brought back online.

Monitoring network connectivity with ESXTOP

Like CPU, memory, and disk, ESXTOP can monitor network connectivity of your physical NICS (uplinks) and virtual NICs associated with VMs. You can see typical data transmission statistics; in the case of ESXTOP, this is tracked in MegaBits sent and received per second in the MbTX/s (transmitted/sent) and MbRX/s (received). These statistics can be useful to understand the existing traffic and whether a physical NIC can continue to support additional VMs or whether a particular VM is producing excessive amounts of traffic.

Two statistics that can immediately show whether a physical NIC is overcommitted or even connected to a faulty physical switch port or cable is %DRPTX and %DRPRX. On a physical NIC, you should expect these values to be at or near 0.00. On a VM, %DRPRX could also indicate a VM does not have sufficient CPU, memory, or disk I/O resources; though more typically, CPU.

Summary

In this chapter, we looked how assigning resources to VMs affect all of the resources on the host as well as other VMs. In addition, we looked at the features implemented by VMware vSphere to manage overcommitment such as the CPU scheduler, transparent page sharing, and ballooning. We then discussed storage and the effect drive type and RAID selection has on performance; reviewed important ESXTOP statistics for each of them and wrapped up an overview of networking, typical practices with uplinks and vSwitches, and finally, tips on how to monitor network connectivity. Next, let's look at the advanced settings that help further manage and provide availability to VMs and their resources.

3
Advanced Resource Management Features

Up until now, we have looked at what is needed to run VMware vSphere and how resources are consumed by VMs. In this chapter, we will look at some of the advanced features available to ensure VMs have access to the resources necessary to run properly as well as features that ensure VMs remain accessible.

We will cover the followings topics in this chapter:

- Understanding CPU power management
- Reservations, limits, and shares
- Resource pools and vApps
- vMotion, EVC, and Storage vMotion
- HA, DRS, and App HA
- Affinity and anti-affinity rules
- Fault Tolerance
- Hot Add
- SDRS and datastore clusters
- vSphere Flash Read Cache
- Distributed switches
- Network and Storage I/O Control
- Host profiles and auto deployment
- VSAN

Understanding CPU power management

CPU power management is less of a vSphere feature and more of a feature available on certain CPUs. These power management features allow the CPU to enter a low power state where it does not provide the full capacity of the processor resources to the OS based on demand. When demand increases, it comes out of the lower power state. However, having a feature like this enabled on an ESXi host can affect the performance of your VMs.

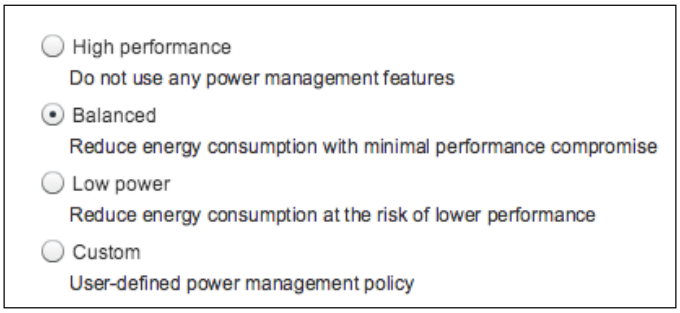
ESXi expects a certain amount of processing resources based on the identified processor and number of cores. If the CPU power management features are enabled, VMs will not receive the full amount of resources that potentially cause an increase in **CPU %RDY**, thus degrading performance. This is because the vCPU expects more resources than are available.

Within ESXi, you can configure the following preferred power management policy settings:

- **High performance**
- **Balanced**
- **Lower power**
- **Custom**

If the system shows a not supported message, then the hardware does not allow power management from the operating system and you will need to configure it as per the vendor's documentation. Except for fringe use cases, you should apply the **High performance** setting to avoid any potential performance problems. To check or change the **Power Management** settings, log in to the vSphere Web Client and navigate to **vCenter | Host and Clusters**, select the host, and click on the **Manage** tab. Under the **Hardware** section, click on **Power Management**.

Here, you can see the default setting of **Balanced** as well as the other options we listed earlier:

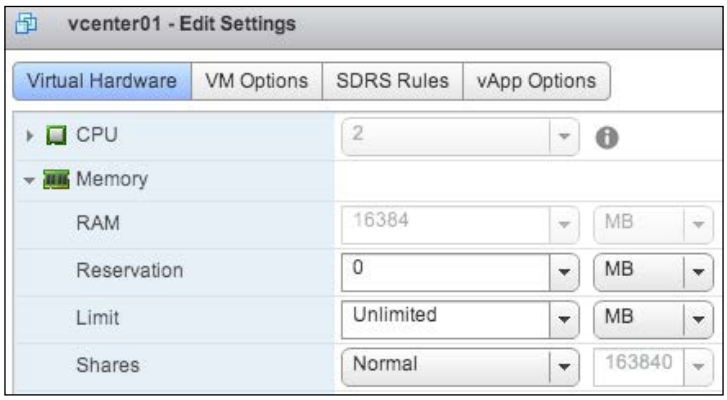


VMware has several useful resources on **Power Management**, including the following:

- KB article 1018206 (`http://kb.vmware.com/kb/1018206`)
- *VMware VROOM! Blog* by *Rebecca Grider* (`http://blogs.vmware.com/performance/2013/05/power-management-and-performance-in-esxi-5-1.html`)
- *Host Power Management in VMware vSphere 5*, a technical white paper (`http://www.vmware.com/files/pdf/hpm-perf-vsphere5.pdf`)

Reservations, limits, and shares

Reservations act very much like you would expect them to—they reserve a certain amount of CPU memory resources to a VM. This can ensure your VM always has access to a specific amount of processor MHz or physical memory. Setting a reservation on a per-VM basis is very straightforward. While logged into the vSphere Web Client, right-click on any VM and select **Edit Settings**. While on the **Virtual Hardware** tab, expand either the **CPU** or **Memory** section. In the following screenshot, you can see the **Reservation** value for this particular VM is set to `0` MB:



As we discussed in the previous chapter, just because we assign 16 GB of memory to a VM does not mean it always has access to 16 GB of physical memory. During times of resource contention, ESXi may swap memory to disk or request memory be returned via the **VMware Tools** balloon driver. Reservations prevent this from happening and ensure the VM always has access to the amount of physical memory or CPU specified in the reservation. Reservations, however, have an impact beyond just ensuring a VM is guaranteed a specific amount CPU or memory. Later in this chapter, we will discuss the admission control that is enabled along with HA.

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Resource limits

Limits act as the opposite of reservations and cap the amount of resources a VM can consume. As we saw in the previous screenshot, limits are configured in the same place as reservations, at the VM level. When you assign a limit to a VM, either on **CPU** or **Memory**, the guest OS is unaware of the limit at the hypervisor level. Using the VM as an example again from the previous section, if we limit our VM (which has 16 GB assigned) to only 8 GB of memory, the guest OS would only have access to 8 GB of physical memory from the host, even though it believes it has 16 GB. ESXi would then have to use its various memory reclamation techniques, as if there was memory contention on the host, such as ballooning or swap to disk, to fulfill the request from the guest OS.

You should also be aware that setting a CPU limit is further limited by the number of vCPUs assigned to the host. For example, if we had a VM with 2 vCPUs and set a limit of 2000 MHz, each vCPU would only have access to 1000 MHz of compute resources from any core. One of the few use cases I have found for limits is in development, test, or quality assurance-type environments. Take an application that, for example, requires 8 GB of memory to install, but because this is a test environment, you know it will never need more than 4 GB. You could assign 8 GB of memory to the VM and set a limit of 4 GB. Now, when the application is installed in the guest OS, it will see 8 GB of memory but ESXi will never allow it to use more than the 4 GB limit.

Resource shares

Shares, when implemented properly, can provide the best of reservations and limits. Shares allow you to set priority for resource access, but this is only enacted during times of resource contention. Let's combine the last two scenarios: you have two VMs, a production VM that is assigned 16 GB of memory, a test VM that is assigned 8 GB of memory, and your host with 20 GB of memory. Under normal circumstances, the host can likely manage the memory between both VMs. If the test VM was under increased load, it could cause performance problems for the production VM. By assigning shares for memory to both VMs, you can ensure the production VM always has access to memory resources over the test VM.

Shares between two VMs are easy to conceptualize and plan; however, in large environments, they need careful planning and monitoring. Imagine you assign memory shares to two resource pools, each with 10 VMs. A resource pool A has 20,000 shares assigned for production VMs and a resource pool B has 10,000 shares assigned for testing VMs within the pool. Six months later, it's determined that the resource pool B should have five of the 10 VMs shutdown, but the shares are not changed. By not changing the shares for the resource pool B, the VMs will have access to an equal amount of resources as the VMs in the resource pool A, because the 20,000 shares in resource pool A are divided across 10 VMs. However, the 10,000 shares in resource pool B's shares are now only divided between five VMs.

Resource pools

While reservations, limits, and shares can all be set at the VM level, it's much easier to manage them at the resource pool level by grouping several VMs together in some meaningful way; for example, all web servers for a particular application or all VMs responsible for running a particular application. Before we go on, resource pools require DRS to be enabled on the cluster. So, if you aren't licensed for either Enterprise or Enterprise Plus, you won't be able to use resource pools.

As we discussed previously, resource pools must be carefully planned and managed; otherwise, the share settings could be restricting resources to key VMs during times of contention (recall that shares are only used if there is resource contention unlike limits and reservations).

vApps

vApps is another feature that, like resource pools, requires DRS, so you will again need an Enterprise or Enterprise Plus license. Similar to resource pools, you can use vApps to assign reservations, limits, and shares to groups of VMs. One of the more useful features of vApps is that you can control the power on order of all VMs within a vApp. Let's look at an example to make more sense of this.

Consider a basic two-tier web application that has a web server and database server. In order for the application on the web server to function properly, the database server must be powered on and accept connections. If both of these VMs were placed into a vApp, you could power on the vApp as a single command, which could first power on the database VM, then after a certain period of time, power on the web server. While it's not a foolproof way to guarantee all servers have started properly, it's another tool provided by VMware vSphere.

vApps also include settings to control IP assignment for VMs within the vApp, providing either DHCP, static, or Transient IP allocation from a vCenter IP Pool. IP pools, however, is an advanced feature that requires developers to use VMware Studio to create vApp in order to work properly. IP Pools are not a replacement for either DHCP or manual IP assignment.

vMotion

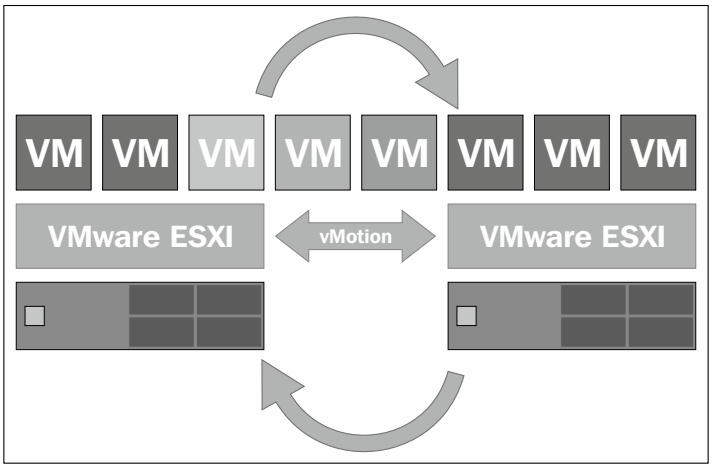
vMotion is the feature that puts VM and virtualization as a whole on the map. vMotion is the process of migrating a VM from one host to another while it is running with no downtime. If you have ever had to maintain hardware and perform maintenance that required the physical server to be shut down, you can already imagine what an amazing tool this is.

vMotion is included even with the Essentials Plus and Standard license kits, so this feature is available to most licensed customers. The only license kit that vMotion is not included in the vSphere Essentials kit, which is essentially just a license for vCenter to manage up to three ESXi servers each with up to two processors.

To vMotion a VM, you must have at least two hosts managed by vCenter with at least one VMkernel interface enabled for vMotion traffic. Both hosts must have compatible CPUs from the same vendor; you cannot vMotion a VM from a host with an Intel processor to a host with an AMD processor. As of vSphere 5.1, you are no longer required to have shared storage as it can migrate both the VM state and storage in the same process. We will discuss storage vMotion shortly, so for now, I will focus on just the migration of the VM state.

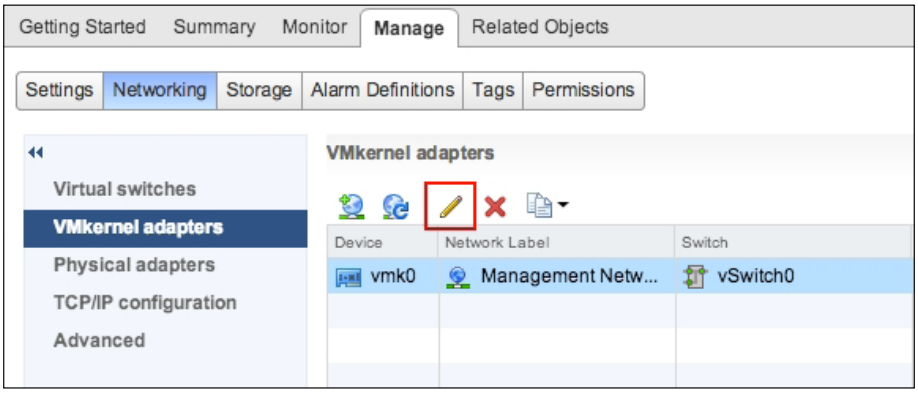
An initial copy of the VM state is made on the destination host. During this process, the VM is still running on the original host, and the copy running on the new host is not responding to requests yet. The initial pass does take some time to complete. So, in order to complete, the vMotion process will continue to make passes, copying any memory changes since the last copy started, eventually getting to a point where the state of the VM is completely copied to the second host. Once complete, the VM on the original host is stunned to give the destination host time to start managing the VM and respond to requests.

In the following figure, a VM residing on one host is copied to another host. The copy process repeats until the VM's memory state is completely copied to the new host:



The amount of time it takes to fully migrate the VM will depend on your network, the speed of your NICs, and switches, as well as the amount of memory assigned to the VM. As we have discussed several times, overcommitting resources to a VM is not recommended.

Typically, vMotion traffic is set up using a dedicated physical uplink so that it does not interfere with the management or VM network traffic. However, it can be enabled over the default VMkernel (also known as management) port group, though this is typically not considered a good practice. To enable vMotion, log in to the vSphere Web Client and navigate to **vCenter | Hosts and Clusters**. Then, select the host you wish to enable vMotion on, then go to the **Manage** tab and click on **Networking**. Now, click on the **VMkernel adapters** that you wish to run vMotion traffic over, then click on the pencil icon as shown in the following screenshot:



If you have not created additional **VMkernel adapters**, then you will most likely only see the one that is marked for management traffic created during the ESXi installation. Now, check the box next to vMotion traffic and you are ready to start migrating VMs. While vMotion is very easy to enable, like any other feature, it requires careful planning against your business requirements and network environment to ensure vMotion can complete in a timely manner.

In vSphere 5.1, a new feature called **Multi-NIC vMotion** was introduced. This allowed you to use several physical NICs to migrate VMs more quickly, even if they were not bonded together and connected via a supported switch configuration such as LACP.

Enhanced vMotion Capability (EVC)

Enhanced vMotion Capability or EVC was introduced with ESX/ESXi 3.5 and allows you to mix and match servers with different CPU versions. You are still, however, required to have CPUs from the same manufacturer; you cannot vMotion a powered on VM from a host with an Intel CPU to one with an AMD CPU.

Using the VMware compatibility guide that we used in *Chapter 1*, *Understanding vSphere System Requirements*, you can also search for CPUs compatible with a specific EVC mode. For example, if we search for all CPUs compatible with the Intel Ivy-Bridge Generation, we will be a given a list of all CPUs that can be used in that EVC mode, as seen in the following screenshot (note that not all CPUs are shown):

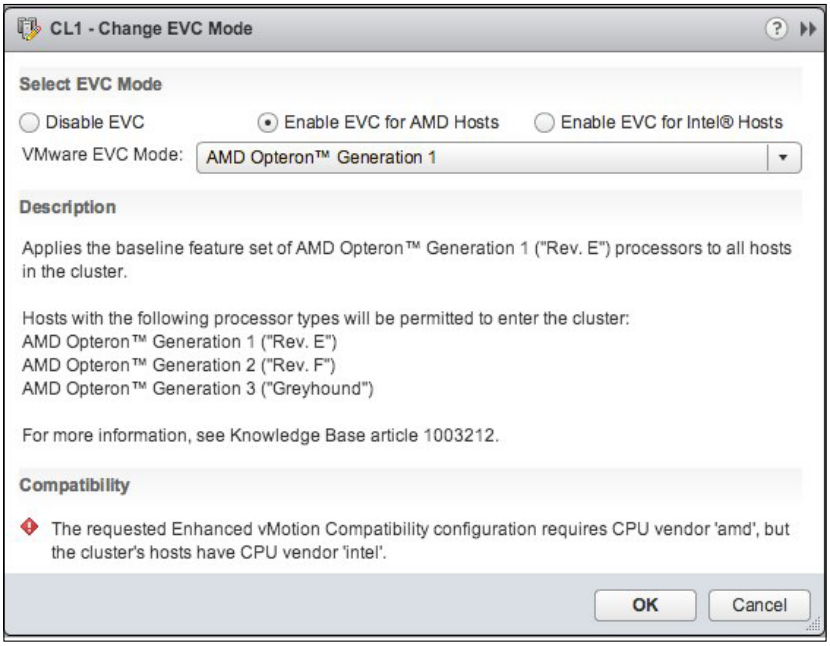
EVC Modes	Intel® Ivy-Bridge Generation
Intel Xeon E3-1200-v2 Series	✓
Intel Xeon E5-2600-v2 Series	✓
Intel Xeon E3-1200-v3 Series	✓

EVC plays an important role in the ongoing life cycle of a cluster. Imagine a scenario where 10 hosts can adequately provide resources for all of the VMs in an environment. As the organization grows, you may need to add additional hosts to provide more capacity, but the CPUs used in the initial deployment may no longer be available. EVC allows you to buy new hosts and integrate them into your existing cluster and processes.

How to enable EVC

Enabling EVC is not difficult; however, you may need to power off existing VMs in the host if the CPU has a feature set newer than the EVC mode you are enabling. This is because those features from the newer CPU will no longer be available when the EVC is enabled. To enable EVC, log into vSphere Web Client and navigate to **vCenter | Hosts and Clusters**. Select the cluster you wish to enable EVC on and click on the **Manage** tab. Select **VMware EVC** and click on the **Edit** button. You can now choose either the EVC mode for AMD or the EVC mode for Intel hosts and then the appropriate EVC mode.

Once you select the EVC type you wish to use, vCenter will check the hosts to ensure they are compatible. For example, in the following screenshot, **EVC for AMD Hosts** has been selected, which we cannot use because the CPUs in the hosts are Intel-based.



You will be required to enable EVC if your cluster contains hosts with different CPU models and has DRS enabled. Once EVC is enabled, it masks features in the CPU that are not compatible with that mode. For example, if you set the EVC Mode to **AMD Opteron Generation 1**, you would not have access to features in Opteron Generation 2, even if other hosts use that type of processors. For more information on VMware EVC modes, refer to the KB article 1003212 at `http://kb.vmware.com/kb/1003212`.

DRS

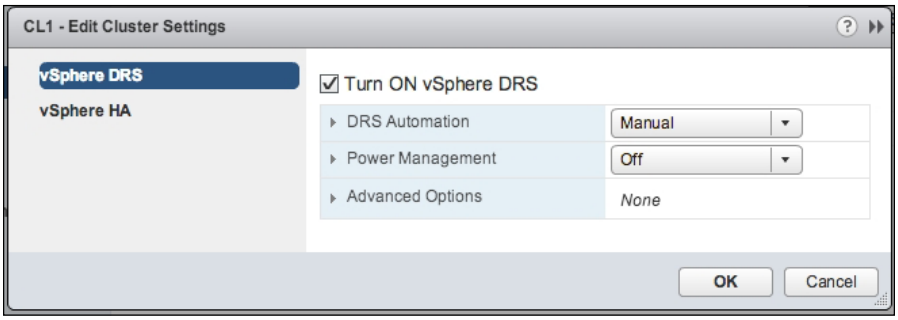
Both DRS and HA are extremely advanced topics. In fact, one of my all-time favorite books, *VMware vSphere 5.1 Clustering Deepdive*, *Duncan Epping* and *Frank Denneman* is devoted entirely to this subject, and I'd highly suggest you pick up a copy. Because of how deep the book dives, it takes a few reads, but it is well worth the time investment. Because of the value this book has provided me, I am honestly not sure I could write this section without thanking both authors of the book.

DRS builds up on your vMotion configuration. Whereas vMotion allows you to manually relocate VMs, DRS provides automated migration and initial placement of VMs based on the workload to ensure resources are readily available to your VMs. Additionally, DRS uses a feature called **Distributed Power Management** (**DPM**) to automatically power off unnecessary hosts to conserve power.

Enabling DRS is quite straightforward. Like EVC, it's enabled at the cluster level, but you do have to consider the impact on your environment. For example, since DRS uses vMotion to help balance the workload by migrating VMs to host with available resources, you'll need to consider the maximum number of concurrent vMotion operations per vMotion network, whether its 1 Gbps or 10 Gbps, the maximum number of operations per datastore, and so on.

DRS-enabled clusters can only have a host with the same type of processor, either Intel or AMD, and EVC configured if there are different CPU models to ensure vMotion will work as required. Each host in the cluster will also need access to a shared datastore(s) so that each host has access to all of the VM-related files such as the VMDK (the virtual hard disk), VMX (the configuration file), and swap file. You'll need to consider the swap file location if you have moved them out of a shared datastore, for example, if you have configured them to be stored on the local server flash storage.

To enable DRS while in vSphere Web Client, navigate to **vCenter | Hosts and Clusters**. Select the cluster you wish to enable DRS on and click on the **Manage** tab. Under Services, select **vSphere DRS** and click on the **Edit** button. Check the box next to **Turn ON vSphere DRS** to configure the settings.



There are three automation levels you can choose to enable for DRS: **Manual**, **Partially Automated**, and **Fully Automated**.

- **Manual**: This DRS will make suggestions as to where to place new VMs and make suggestions for VM relocation when requested
- **Partially Automated**: This DRS will automatically place newly created and VMs being powered on and make suggestions for VM relocation when requested
- **Fully Automated**: This DRS will automatically manage the placement and relocation of VMs

DRS, while it is a fantastic feature, introduced an interesting problem. What if you have a clustered/highly available application composed of multiple VMs and DRS places and all of these VMs on a single host? All of the application-level configuration and planning would be for nothing if that single host were to go down. To address this, VMware uses affinity and anti-affinity rules.

DRS affinity and anti-affinity rules

There are two types of affinity and anti-affinity rules: host-based and VM-based. Host-based rules ensure a VM is either on (affinity) or not on (anti-affinity) a specific host. If we consider the example we discussed at the end of the DRS section, you could create a rule to ensure that VM-A is placed on Host-A, while VM-B is placed on Host-B. Now, if one of the two hosts goes offline, at least one of the VMs will still be running (assuming no previous action to migrate the VM was taken). An anti-affinity host rule would prevent a VM from being placed onto a specific host.

You can also create affinity and anti-affinity rules on a VM-to-VM basis, which in my experience is a bit more useful. Rather than creating a rule that says a VM can or cannot be on a certain host, you can create rules that group VMs together on the same host (affinity) or ensure the VMs are running on different hosts (anti-affinity).

It should be noted that affinity and anti-affinity rules are not obeyed by VMware **High Availability** (**HA**).

High Availability

While DRS is a feature that allows the automatic placement of new and running VMs to hosts, which can ensure resource requirements are met, VMware vSphere High Availability (or HA as it's commonly referred to) is a feature that helps restore VMs to a running state in the event of an unplanned host failure.

It's important to understand that HA is for unplanned outages, such as a component failure in a host. VMs running on a host that suffers an unplanned outage are shutdown when the host fails and are powered up on an available host within the cluster.

While HA is enabled at the cluster level like DRS, it functions exclusively at the host level, so a failure in vCenter will not prevent HA from working normally. In fact, HA is a great way to ensure vCenter uptime since a clustered or load balanced version of the vCenter application is not supported. For more information on support vCenter HA options, see VMware KB 1024051 at `http://kb.vmware.com/selfservice/microsites/search.do?language=en_US&cmd=displayKC&externalId=1024051`.

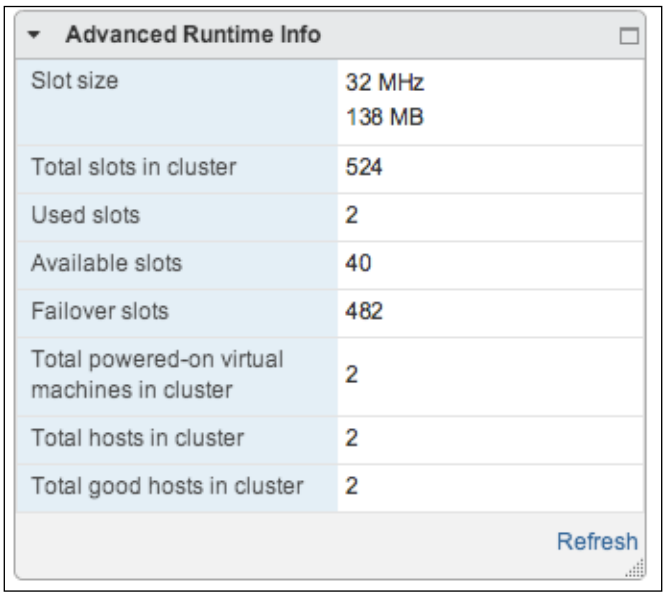
When HA is enabled, an agent is installed on each host. This agent is responsible for communication between all the hosts to ensure the hosts are online and running as expected. As of vSphere 5, the agents operate in a master/slave mode. A master is selected from the hosts and the other hosts act as slaves. HA will then create a file on the shared datastores in a folder called `vSphere-HA` with an inventory of all VMs and their power state.

HA communicates over the VMkernel interface for traffic management. However, it also uses shared datastores to monitor host availability in the event the management network is not available. As we discussed previously, it's quite common, if not best practice, to isolate various types of traffic related to your VMware vSphere environment; the examples we looked at were management, vMotion, and general VM network traffic. If your management network were to become unavailable, HA would verify through the selected datastores whether or not the VMs that belong to that host were still running.

Admission control

One feature of HA that is often misunderstood is admission control. Admission control ensures that your cluster has enough computing resources to satisfy all of the running VMs. There are three admission control policy options that you can select from based on your specific requirements.

The default option, **Define failover capacity by static number of hosts**, ensures that the cluster has enough resources for the specified number of hosts to fail. To do this, it calculates a slot size based on the CPU and memory reservation of all the powered on VMs and then calculates how many slots each host can support. With this option configured, you can see how many slots your cluster has available and the number of used slots. Once logged into vSphere Web Client, navigate to **vCenter | Hosts and Clusters**. Select the cluster you wish to view the slot sizes for and click on the **Monitor** tab. In the vSphere HA menu, you will see **Advanced Runtime Info**. This section shows you the slot being used for the calculation, the number of slots available, and the number of slots in use, as seen in the following screenshot:



Advanced Runtime Info

Slot size	32 MHz 138 MB
Total slots in cluster	524
Used slots	2
Available slots	40
Failover slots	482
Total powered-on virtual machines in cluster	2
Total hosts in cluster	2
Total good hosts in cluster	2

Refresh

In this cluster, where there are no reservations set, you can see that it's only using 32 MHz and 138 MB to calculate the slot size. While this will most likely ensure you can power on many VMs within the cluster, it also means you may not realistically be able to maintain an adequate amount of resources for all of the VMs running in the event of a host failure.

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If, on the other hand, you have even one VM with a large reservation, you could alter the percentages negatively where you only have very few slots available. Let's look at the same cluster but after assigning a 1024 MB reservation for one of the VMs. As you can see, the slot size is now based on the 1024 MB limit plus memory overhead.

Advanced Runtime Info

Slot size	32 MHz 1162 MB
Total slots in cluster	112
Used slots	2
Available slots	3
Failover slots	107
Total powered-on virtual machines in cluster	2
Total hosts in cluster	2
Total good hosts in cluster	2

Refresh

You may be wondering why the **Total slots in cluster** field does not match the available slots plus used slots. In this case, the hosts are not well balanced. One host has 8 GB of memory, while the other has 128 GB of memory. Combining this, based on the slot size calculation, we have 112 slots in the cluster. However, if the host with 128 GB of memory were to fail, we would only be left with a host which has 8 GB of memory. The other option available for configuring a slot size is to manually specify it. This is useful if you have a few VMs with very high reservations that might otherwise impact your total number of available slots.

You can also configure HA to reserve a specific amount of resources in the cluster. Doing this requires a detailed understanding of your environment so that you can properly calculate what percentages you will need for CPU and memory to support your specific applications and workloads.

First, you need to calculate how many compute and memory resources you have in the cluster. Let's use CPU/compute as an example to determine the appropriate settings for your environment.

If you have 10 hosts, each with 10 GHz of total compute, then 100 GHz would represent 100 percent of the total compute resources. Now, you need to determine how many compute resources all of the running VMs require. If all of your VMs require 25 GHz, that would leave 75 percent of the cluster resources to CPU failover capacity (100 GHz-25 GHz/100 GHz). If you set the **Reserved failover CPU capacity** setting to 25 percent and are using 25 percent of those resources already for current VMs, then admission control will allow you to use up to 50 percent, or 50 GHz of compute resources in this case. This is before admission control would prevent the power on of new VMs. The same calculation would be used to determine an appropriate **Reserved failover Memory capacity** setting.

The final admission control option is **Use dedicated failover hosts**. With this setting, admission control will keep the specified hosts completely idle, reserved just for the failover capacity. While this is a fairly safe option, it also works against leveraging the capacity of all the hosts in your cluster.

App HA

App HA is a new feature in vSphere 5.5, which brings application level restarts for critical applications. Like HA for VMs, App HA restarts applications based on the configuration and applications you have monitored. Since App HA is a new feature, the applications supported are slightly limited but certainly useful for most modern applications.

The applications supported at the time of writing this book include the following:

- Apache Tomcat 6.0 and 7.0
- IIS 6, 7, and 8
- Microsoft SQL 2005, 2008, 2008 R2, and 2012
- Apache HTTP 2.2
- SharePoint 2007 and 2010
- SprinceSource tc Runtime 6.0 and 7.0

The requirements are ESXi and vCenter 5.5 as well as vCenter Hyperic (a monitoring solution provided by VMware). Assuming you meet these requirements, App HA is deployed as a virtual appliance via the OVA provided (an OVA is simply a packaged VM that is platform agnostic, meaning you could import the OVA into ESXi, or Citrix Xen Server, and so on). Because App HA requires Hyperic, a feature outside the scope of this book, we will not go into the details of how to configure it; however, you can find more information in the *vSphere App HA Documentation Center* at `http://pubs.vmware.com/appha-1/index.jsp`.

Fault Tolerance

Fault Tolerance, included with standard and above licensing, is one of the more powerful features of vSphere but also possibly one of the least used at the same time. Fault Tolerance creates a copy of the protected VM and keeps it in sync during normal operation. If there is a failure to the host running the protected VM, the copy can immediately take over as if the original VM never went down. While FT is enabled at the VM level, it's important to note that it protects against host failure like HA does, but does not require the VM to restart like HA. After the secondary copy becomes active, HA will boot a new secondary VM to ensure FT is able to resume.

Because the secondary copy is kept in sync, it is unable to protect against an application failure. When an application within the protected VM fails, the application also fails on the secondary VM.

Fault Tolerance has the following specific requirements:

- The protected VM must be in an HA-enabled cluster.
- The hosts must have access to the same shared storage location that the protected VM is stored on.
- A VMkernel adapter must be configured for vMotion and Fault Tolerance logging. They can be on the same adapter, but if you are making heavy use of FT, you should consider a dedicated adapter.
- The VM cannot have any unsupported devices connected, such as local USB or CD drives.
- The VM must be configured with a VMDK or virtual mode RDM.
- The VM must only have a single vCPU, as virtual SMP is not supported.

The last bullet point is the largest pitfall of FT at this time. Most applications that might leverage FT are likely to also require multiple vCPUs; however, if your application does not, then FT is a great way to ensure continuous availability. Multi vCPUs/virtual SMPs have been rumored for some time but are not yet available.

For more information on FT, see the *vSphere Availability* guide at `http://pubs.vmware.com/vsphere-55/topic/com.vmware.ICbase/PDF/vsphere-esxi-vcenter-server-55-availability-guide.pdf`.

Hot Add

Hot Add is a feature that allows you to add virtual hardware to a VM while running. This feature, previously present only in advanced editions of vSphere, is now available with standard and above licensing. While this feature is available via VMware vSphere, not all operating systems support adding resources such as CPU or memory while running.

Systems running Data Center Edition of Windows 2008 and above, and most modern Linux kernels support this feature, though you'll need to verify this for your specific edition.

Storage vMotion, Storage DRS, and Datastore clusters

Till now, we have focused on features and practices to ensure CPU and memory resources are available to your VMs. However, there are several features that, like their compute and memory-focused siblings, help to ensure your storage resources are readily available. The storage-focused version of the features we have previously discussed offer much of the same focus.

Feature (Compute/Memory)	Function	Feature (Storage)	Function
vMotion	This moves the CPU and memory from a running VM to another host	Storage vMotion	This moves the virtual hard drive, configuration file, and other related files to another datastore
DRS	This provides recommendations on placement and can automatically move VMs to a host that can support CPU and memory requirements	Storage DRS	This provides recommendations on virtual hard drive and other related files and can automatically move storage-related files to another datastore with adequate available space.
Clusters	This creates a logical group of compute resources that can be shared among VM through the use of vMotion and DRS	Datastore cluster	This creates a logical group of datastore resources which can be shared among VMs through the use of vMotion and DRS

Storage vMotion

Storage vMotion provides administrators with the ability to migrate VM storage to a new datastore while running. This allows you to manually balance space on your datastores or migrate a VM to a different tier of storage based on the new requirements. Let's consider a couple of examples. Storage vMotion, as of vSphere 5.5, has been included with standard and above licensing. Prior to 5.5, this was only available in the Enterprise/Advanced and higher level licenses.

As more VMs are added to your environment, the space available on your datastore will decrease. Because of the increase in datastore utilization, you decide to add a new group of drives to your storage appliance, create a LUN, and present the LUN to your ESXi hosts as a new datastore. Storage vMotion, in this scenario, would allow you to relocate all of the files for a VM to the newly added datastore. Another option, as you might be thinking, would be to expand the datastore at the storage appliance level. While this is certainly possible, it would depend on your specific storage appliance and the impact it imposes on your datastore, such as decreased I/O performance while it adds the new drives. In some cases, you may decide to just add the additional datastore. Also, as we discussed in *Chapter 2*, *Assigning Resources to VMs*, most storage appliance vendors may have recommended array sizes, which you would not want to exceed so that you are not overloading the management or storage controllers in the appliance.

Now let's walk through a performance-related scenario. A VM is created for testing a specific application on a datastore with a low IOPS profile such as the one created on SATA drives. After testing, you decide to promote the VM into production for all users to access. With additional users accessing the application, there will be additional IOPS required, which the SATA-based datastore cannot meet. Here, you could migrate the VM storage to a datastore created on flash storage to meet the new IOPS requirements.

Without these features or on a physical server bound to specific hardware, this would have been much more difficult to manage without incurring some form of downtime. With Storage vMotion, you can do this while the VM is running and being accessed.

Unlike vMotion, there are no additional networking requirements to set. The vMotion process is managed by the ESXi host unless your storage appliance supports the VAAI XCOPY primitive (`http://www.vmware.com/files/pdf/techpaper/VMware-vSphere-Storage-API-Array-Integration.pdf`). As we discussed in *Chapter 2*, *Assigning Resources to VMs*, if you have a storage appliance that supports VAAI, it can handle requests that would otherwise need host CPU and memory resources to complete.

Datastore clusters

Datastore clusters, just like compute clusters, are a logical representation of resources presented to your hosts. A datastore cluster is made up of independent datastores. Datastore clusters, similar to how a compute cluster requires CPUs from the same manufacturer, must be made up of the same type of datastores with the following considerations:

- A datastore cluster must either be a VMFS datastore (FC or iSCSI) or NFS; you cannot combine VMFS and NFS datastores into the same datastore cluster.
- You can only attach ESXi 5.x hosts to the datastore if it will be added to a cluster. If there are any hosts running 4.x or 3.x, you will need to either upgrade the host or disconnect it from the datastore.

To create a datastore cluster, navigate to **vCenter | Storage** in vCenter Web Client, click on the **Actions** button, and select **New Datastore Cluster**. The **New Datastore Cluster** wizard will begin by creating a compute cluster; name the datastore cluster, and select whether you would like to **Turn ON Storage DRS**, which we will discuss shortly. The steps for creating a datastore cluster are as follows:

1. Select the automation level for Storage DRS; you will be asked to choose this setting whether or not you selected to turn on Storage DRS on the previous page. These settings will then be saved if you decide to enable Storage DRS later.
2. Next, select whether you want to enable I/O Metric Inclusion, which will factor I/O latency into Storage DRS recommendations. On this tab, you can also select the storage utilization threshold you wish to attain on each of the datastores.
3. If you expand **Advanced Options** in step 2, you can choose the default VM affinity rule, which is to keep VMDKs together. If you do not select this option, then DRS may place different VMDKs for the same VM on different datastores. That's not necessarily a bad thing, just a design decision you need to make based on your VM and application requirements.
4. Select the compute cluster you wish to enable this datastore cluster in.
5. Next, select the datastores you wish to include. Datastores will need to be available on all hosts in the cluster, otherwise you will see a warning that reads **Host Connections Missing**.
6. Finally, you can review the settings before clicking on the **Finish** button.

Once you click on **Finish**, the datastore cluster will be created and any datastores selected will be placed into the cluster. While you can select local disks to be part of the cluster, carefully consider the impact on availability. If the host fails, there would be no way for the HA to restart the VMs on a new host. You can see a complete datastore cluster in the following screenshot:

While a datastore cluster can be created with a Standard vSphere license, the real automation comes from Storage DRS, which requires at least one datastore cluster to be configured and a vSphere Enterprise Plus license.

Storage DRS

While Storage vMotion is available with the vSphere Standard license now, Storage DRS still requires an Enterprise Plus license. If Storage vMotion is the tool for manual migrations and load balancing, then like DRS is for compute and memory, Storage DRS is a tool designed to automate the placement and migration of VM storage to a datastore capable of providing the resources needed by the VM.

Storage DRS has the following two modes:

- **No Automation** (manual mode): This provides recommendations that you will need to choose to apply.
- **Fully Automated**: This will make the initial placement and migrations for you. Placement and migration is based on two factors: space utilization and I/O latency.

Storage DRS employs the concept of affinity rules to control the placement of VM storage on specific datastores. As we saw during the datastore cluster creation, you have the option of enabling the default VM affinity rule to keep VMDKS together, or you can disable this setting, which would allow Storage DRS to select different datastores for different VMDK files belonging to the same VM.

Storage DRS automation levels as well as affinity rules can also be set on a per VM basis. To do this, click on the **Settings** tab with the datastore cluster selected and click on **VM Overrides**. The **VM Overrides** section allows you to add specific VMs that will use different rules than the defaults for the cluster. For example, maybe you have a VM that must always run on a datastore backed by flash storage. Even if your Storage DRS automation settings are fully manual and the affinity rule is not set to keep VMDKs together, you could add the specific VM to the VM overrides section so that Storage DRS does not take any action on that VM.

One important thing to note when comparing DRS and Storage DRS is that DRS operates in real time, migrating VMs from host to host; however, Storage DRS operates at a set time interval (which you can change), that is, every eight hours by default. We can set the interval in the advanced settings of the datastore cluster.

vSphere Distributed Switches

We briefly mentioned distributed switches in *Chapter 2*, *Assigning Resources to VMs*. If you recall, a virtual switch acts in the same fashion as a physical switch; VMs connect to the switch or port group that controls network access through the use of VLANs. The virtual switch then uses a physical NIC as an uplink to a physical switch so that VMs can communicate with VMs on a different host and other devices connected to the network.

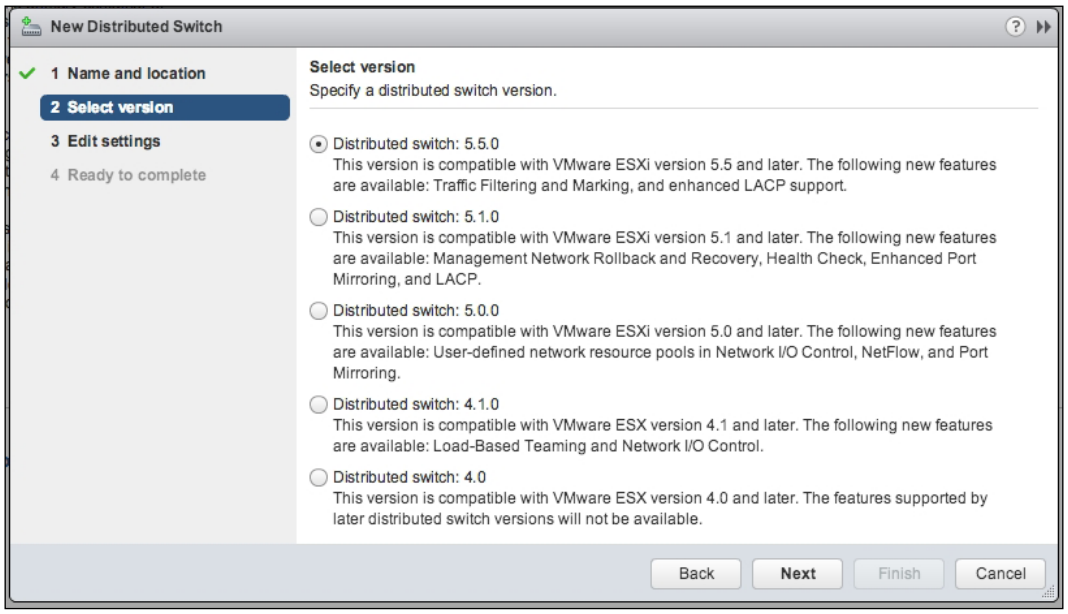
vSphere Distributed Switches (vDS) is an advanced version of vSwitches available with Enterprise Plus licenses. A vDS offers several advantages over standard vSwitches (vSS), including the following:

- Easier management across all hosts connected to the vDS; a change to the vDS is applied to all hosts whereas a change on a vSS has to be performed on each host
- Support for private VLANs
- Inbound traffic shaping
- **Network IO Control** (**NIOC**) that allows network bandwidth to be allocated to network resource pools using either predefined traffic types (VM network traffic, vMotion traffic, and so on) or custom-defined traffic types

A vDS operates like a modular physical switch. vCenter provides the management features, similar to what a supervisor card in a Cisco switch would do, and provides its configuration to each of the host, which then act in a similar fashion to a line card, that is, receiving its configuration from the supervisor.

There are two types of vDSs: the native VMware vDS and third-party vDSs such as the Cisco Nexus 1000v. In this section, we will focus on the native VMware vDS. If you use or need to use a third-party vDS, refer to the documentation from that vendor.

To create a vDS, log in to the vSphere Web Client, navigate to **vCenter | Networking**, and click on the **Getting Started** tab. At the bottom of the page, click on the link labeled **Create a distributed switch**. After naming your vDS, you will be given the option to select the vDS version; this allows backwards compatibility with other hosts in the cluster that may not be on the same version. Each version will specify the features available above and beyond the previous version. For example, with a vDS 5.5, there is an enhanced LACP support that is not available in a vDS 5.1.



Once you have selected the version, select how many uplinks should be added to the vDS, and whether to enable **Network I/O Control** and the name of the default port group. Once the vDS wizard finishes, the final step is to add hosts to the vDS and migrate any existing VMs.

Network I/O Control enables resource pools to separate various types of traffic, ensuring they are provided a share of the available network resources. You set shares at the vDS level using the **Resource Application** tab. By default, NIOC adds several predefined types of traffic; however, you are free to add resource pools for your specific use cases as well. You can also set limits on a network resource pool, limiting the available amount of bandwidth available to it. As we discussed with limits earlier, limits are always in place, even if there is no contention, so shares are the preferred method of managing NIOC.

Select the newly created vDS and click on the **Add and manage hosts** link. Since this is a new switch, we will need to first click on **Add hosts** to add hosts to the vDS. On the **select hosts** screen, click on the **Add** button to do this. If all of your hosts do not appear, click on the **Incompatible Hosts** link at the top of the window. You will see incompatible hosts if any here. For example, if they are not licensed for Enterprise Plus or if they are running a version of ESXi not compatible with the vDS version you selected.

Once the hosts are added, click on **next** to select the options you wish to select. By default, it will add physical adapters and VMkernel adapters to the vDS. If you already have a vSS running with VMs connected, you can also select **Migrate virtual machine networking** to move the VM NICs to the vDS.

Select the adapters you wish to use for uplinks, and click on the **Assign Uplink** button to map the physical NIC to the vDS uplink. Next, you can choose whether to migrate the existing VMkernel adapters or create a new one. If you choose to create a new one, a new wizard will launch that will guide you through that process; otherwise, select the existing VMkernel adapter and click on the **Assign port group** button.

Next, you can choose which VMs you want to migrate to the vDS. Select the VM and click on the **Assign port group** button to select the desired port group. Finally, click on **finish** to add the host and any VMs to the vDS. While this sounds a bit scary, and it's certainly something you should test in your lab, VMware added a new feature in vSphere 5.1 called **vSphere Network Rollback**. If ESXi detects a misconfiguration, the task will fail and revert to the previous state. For more information on Network Rollback, see VMware KB 2032908 at `http://kb.vmware.com/kb/2032908`.

A new feature as of vSphere 5.1 is the ability to import, export, and restore vDS to correct errors. VMware KB 2034602 has more information on this at `http://kb.vmware.com/kb/2034602`. One last option is the ability to restore networking through **Direct Console User Interface** (**DCUI**), the interface you interact with when a keyboard and monitor are directly connected to the ESXi host.

When using a vDS, be sure to have a good recovery plan for vCenter and its database. While each of the hosts will continue to operate if vCenter goes down, you will not be able to edit those vSwitches as all of the configuration is stored in the vCenter database. If you are using Microsoft SQL, see Chris Wahl's blog post on creating SQL maintenance plans at `http://wahlnetwork.com/2012/04/06/protecting-the-vcenter-sql-database-with-maintenance-plan-backups/`.

New in vSphere 5.5

VMware announced two very exciting new features along with vSphere 5.5: vSphere Flash Read Cache and VSAN. Both of these features provide software solutions to common and complex problems in today's modern data center.

vSphere Flash Read Cache

VMware joined the ranks of new vendors such as Infinio, ProximalData, and PernixData who have all created server-side caching systems to improve storage performance by delivering hot data directly from the server instead of having to request it from the SAN.

vSphere Flash Read Cache (**vFRC**), as the name suggests, is a read cache that utilizes flash storage on a host to quickly serve read requests from VMs. The effects vFRC will have on your environment will be based on your workload; however, it stands to reason that any I/O that can be served from a local cache, rather than from your SAN, can help improve overall performance. The SAN does not have to service every read I/O because it is served from cache, and another I/O can process this directly from disk. If you move even just a few hundred IOPS to vFRC, that is a few hundred more, your SAN is able to serve beyond what it could do before vFRC.

Enabling vFRC has only a few requirements as follows:

- ESXi 5.5 is running on the host and is managed by vCenter 5.5
- Hosts are licensed for vSphere Enterprise Plus
- Hosts have local flash storage in your ESXi host that can be dedicated to vFRC (for example, it is not part of another datastore)
- VMs are using hardware compatibility mode v10 (5.5)

To enable vFRC, log in to the vSphere Web Client, navigate to **vCenter | Hosts and Clusters**, click on the ESXi host you wish to manage, and navigate to **Manage | Settings | Virtual Flash Resource Management**. Click on the **Add Capacity** button and select the SSD device(s) you wish to use for vFRC.

Once vFRC is enabled on the host, you have to enable vFRC per VM (VMs do not automatically use VFRC). Right-click on the VM you wish to enable vFRC for and select **Edit Settings**. Click on the **Virtual Hardware** tab and then expand the hard disk(s) you wish to enable vFRC for and enter the amount of cache you wish to provide in the **Virtual Flash Read Cache** textbox.

While vFRC is a new feature, there is already some great content on it from VMware and within the community. For more information, check out the *Performance of vSphere Flash Read Cache in VMware vSphere 5.5* technical white paper at `http://www.vmware.com/files/pdf/techpaper/vfrc-perf-vsphere55.pdf` and the *vSphere Flash Read Cache – Official FAQ* white paper at `http://blogs.vmware.com/vsphere/2013/10/vsphere-flash-read-cache-official-faq.html`.

To see vFRC in action, check out the #*vBrownBag* show by *Nick Marshall* on vFRC at `http://professionalvmware.com/2013/09/vbrownbag-follow-up-nick-marshall-nickmarshall9-covering-vsphere-5-5-vflash-read-cache/`.

As always, the *VMware vSphere 5.5 Documentation Center* will have the most up-to-date information at `http://pubs.vmware.com/vsphere-55/index.jsp#com.vmware.vsphere.storage.doc/GUID-07ADB946-2337-4642-B660-34212F237E71.html`.

VSAN

VSAN is an exciting new technology that was announced at VMworld in August 2013. It is so new in fact that writing this book, it is still in beta stage.

VSAN creates a shared pool of storage from the local storage on each of your ESXi hosts. In order to ensure performance, each host that provides storage must have at least one SSD and for capacity, one HDD (traditional spinning disk). SSDs cannot be shared between VSAN and vFRC, they must be dedicated. An ESXi host can use VSAN storage presented from other hosts even if it is not providing storage to the VSAN cluster.

VSAN is easy to configure and has only a few requirements like vFRC:

- ESXi 5.5 is running on the host and managed by vCenter 5.5
- Each host must provide, at a minimum, 1 SSD and 1 HDD to contribute the storage to the VSAN cluster
- A VMkernel interface marked for Virtual SAN

At this time, it's believed that VSAN, while enabled in vCenter, will be licensed as a separate product and not included with any vSphere licenses, though this could always change in the future.

You can sign up for the beta version at `http://www.vmware.com/vsan-beta-register.html`, and join the VSAN Beta community for news and updates at `https://communities.vmware.com/community/vmtn/vsan`.

VSAN is enabled at the cluster level like HA or DRS. To enable VSAN, log in to vSphere Web Client, navigate to **vCenter | Hosts and Clusters**, right-click on the cluster you wish to enable VSAN for, and click on **Settings**. Under **Virtual SAN**, select **Settings** and click on the **Edit** button.

Check the **Turn ON Virtual SAN** checkbox and choose how to provide disks to VSAN, either automatically or manually

- **Automatically**: This claims all empty disks on each host in the VSAN cluster
- **Manual**: With this, you select which disks are added

Because VSAN is so new, it would be best to use the VMware-provided resources as new information and features can be announced any time. You can learn more about VSAN in the *What's New* section in VMware Virtual SAN's technical white paper at `http://www.vmware.com/files/pdf/products/vsan/VMware_Virtual_SAN_Whats_New.pdf` as well as on the VMware YouTube channel at `http://www.youtube.com/watch?v=cLfbeFoSQaI`.

There is also a #*vBrownBag* podcast on VSAN at `http://professionalvmware.com/2013/09/vbrownbag-follow-up-nick-marshall-nickmarshall9-covering-vsan/`.

Summary

In this chapter, we looked at the various features available with VMware vSphere to not only provide resources to your VMs, but also ensure that your VMs are available even in the event of a host failure. We wrapped up with two new features made available in vSphere 5.5: vSphere Flash Read Cache and VSAN. In the next chapter, we will learn how to automate resource management through several features and tools available with VMware vSphere

4
Automation and Monitoring Options

While VMware vSphere provides several excellent features to enable you to take advantage of the computer, memory, disk, and network resources available, it has also added a new level of scale and management. There are several tools, some licensed and some free from VMware to help you automate the provisioning of new resources as well as monitoring the existing resources in your environment.

In this chapter, we will cover the following topics:

- The automation options available with vSphere for both VMs and hosts
- Examples of how to use the various automation tools to complete common tasks
- Monitoring options available from VMware

Most of the content of this chapter will focus on PowerCLI and the vCenter orchestrator, as they should be the focus going forward. The vSphere CLI, while still made available, is generally used for supporting legacy processes and scripts.

Automation solutions for vSphere

There are several tools that VMware makes available to vSphere administrators. These tools allow administrators to edit hosts or VM configuration settings and create scripts to automate routine tasks such as closing or creating a new VM. Some tools even provide a self-service style portal to end users so they can perform certain tasks without involving the IT/virtualization teams. In the following sections, we will focus on the application/features mentioned in the following table:

Application/feature	Description	Licensed with
PowerCLI	This provides a PowerShell snap-in to manage vSphere resources	This is free. It works with all vSphere editions and the free vSphere Hypervisor (ESXi)
vCenter Orchestrator (vCO)	This is a full-featured automation platform that includes a drag-and-drop interface for creating workflows	This is included in all licensed versions of vSphere and is provided with vCenter on Windows or as a separate appliance

There are, however, many other features that you can use to help automate specific tasks, which are mentioned in the following table:

Application/feature	Description	Licensed with
vSphere CLI (vCLI)	These are command-line utilities built into vSphere Management Appliance (vMA) or installed within an operating system that provide backward compatibility to older ESX commands.	This is free. It works with all vSphere editions and the free vSphere Hypervisor (ESXi)
Templates	This is a vSphere feature that allows you to store common VM types for easy deployment.	This is included under all licensed versions of vSphere
Cloning	This is a vSphere feature that allows you to copy existing VMs to create a new VM with the same settings and state.	This is included under all licensed versions of vSphere

Application/feature	Description	Licensed with
Update Manager	This is a tool for managing updates on ESXi hosts or VMware tools installed on a guest OS.	This is included under all licensed versions of vSphere. It is provided with vCenter on Windows. If you are running the VCSA you will need a Windows server to install update manager.
Host profiles	This is a feature that allows you to specify the configuration options that should be applied to the specified hosts.	This is included under vSphere Enterprise Plus
Auto deploy	This is a feature that allows you to push ESXi to a host during boot time. You can choose a stateless installation, which runs in memory, or a stateful installation, which gets installed to a hard drive. Host profiles can then be used to ensure the configuration is intact.	This is included under vSphere Enterprise Plus

Automation is as much an art as it is a science. While some scripts and tools are likely to work in almost any environment, in most cases, you will need to apply your own business and technical requirements to your scripts and tools. Each of these tools require time to learn, time we don't have here; however, by looking at examples, you can most likely reverse engineer them to fit your needs. Let's look at some common tasks in the various tools. Let's take a quick look at some of the basic features before we get to PowerCLI and vCenter Orchestrator.

Cloning VMs

Cloning allows you to quickly and easily produce new VMs from an existing VM. When you clone an existing VM, it creates an exact copy of the original VM, which you can then reconfigure and power on. During the cloning process, a new MAC address will be provided to any present NIC to prevent the creation of duplicate MAC addresses. Further customization within the operating system would need to be accounted for before you make the clone. For example, in Windows, you may wish to run `sysprep` so it will boot and run through a mini setup to allow you to change the computer name in the operating system. In Linux operating systems, you may need to remove configuration files associated with the NIC to avoid conflicts or problems with NICs not being recognized.

You can create a clone from an existing VM to another VM, clone an existing VM to a template, or clone a template to a new VM.

VM templates

Templates are VMs stored in a different format that is comparable to how a Microsoft Word document template has an extension of `.dotx`, whereas a normal Word document uses `.docx`. Similarly, a normal VM configuration file uses a `.vmx` extension, whereas a template uses a `.vmtx` extension. An example given in the following screenshot shows the files used by a VM template:

Name	Size
vco-template-000001.vmdk	2,790,400 KB
vco-template.vmsd	0 KB
vco-template.vmtx	2.9 KB
vco-template.vmxf	0.26 KB

The VMDK files for a normal VM and a VM you clone or convert into a template will roughly be of the same size. Using the previous example, it's only about 3 MB smaller than the VM we cloned into a template. You can still create new VMs from templates, similar to how you can clone a VM from an existing VM. However, templates cannot be powered, which in theory makes for a more secure baseline image to perform deployments from.

Even though templates cannot be powered on, you can convert a template back to a VM. So you can power it on and make changes to the application or operating system patches, for example; then, you can convert it back to a template. Don't forget any operating system-specific changes you need to make before converting it back to a template.

Another feature available to cloning, either from a VM or a template, is OS customization. OS customization is a vSphere feature similar to an answer file for Windows or a kickstart file for Linux. OS customization settings can be defined in vCenter and selected when deploying a template to provide control over operating system settings such as name, IP settings, and even domain membership. OS customization requires VMware tools to be installed in the VM. The VMware Virtual Machine Admin guide states that you need to have Perl installed for Linux distributions, but Perl is also required to install VMware tools. So it would have already been installed to support the VMware tools installation.

Another form of a template is the **Open Virtualization Format** (**OVF**). If you have deployed any appliance from VMware such as the vCenter appliance, you would have most likely downloaded an **Open Virtualization Appliance** (**OVA**) file, which is the collection of all necessary files to deploy the VM from. You can also create your own OVF/OVA from an existing VM. If you are short on storage space or need to support other hypervisors in your environment, then storing your templates as an OVF may help. The amount of space you save may vary, but in testing, a 75 percent upwards reduction in storage utilization has been observed.

OVFs can also easily be deployed to other hypervisors or to local VMware applications such as VMware Workstation on Windows or VMware Fusion on OSX. You can find the official documentation on cloning, templates, and OS customization in the Virtual Machine Admin Guide available at `http://pubs.vmware.com/vsphere-55/topic/com.vmware.ICbase/PDF/vsphere-esxi-vcenter-server-55-virtual-machine-admin-guide.pdf`.

Update Manager

Update Manager is a tool that allows you to remediate patches for ESXi as well as older ESX versions and perform upgrades. Update Manager maintains information on all of the hosts and downloads updated patches, which can then be pushed to each of the hosts. Compare this to inserting a CD into every ESXi host in your environment, and you can quickly see that Update Manager is a must-have tool for maintaining a secure environment.

Since Update Manger downloads all of the patches needed within your environment, you will need to allocate sufficient amount of storage for patches and the Update Manager database. VMware has published a sizing estimator worksheet at `http://pubs.vmware.com/vsphere-55/topic/com.vmware.ICbase/PDF/vsphere-update-manager-55-sizing-estimator.xls` to help you properly size the Update Manager database and storage requirements giving specific details about your environment. Please note that Update Manager is a Windows-based application. It is not uncommon to find Update Manager installed on the same server as vCenter; however, you should install it on a separate server if possible.

If you are using the vCenter server appliance, you will need to install Update Manager on a separate server.

For more information about Update Manager, see the Update Manager Installation and Administration guide at `http://pubs.vmware.com/vsphere-55/topic/com.vmware.ICbase/PDF/vsphere-update-manager-55-install-administration-guide.pdf`.

Host profiles

Host profiles are a collection of configuration information about your ESXi hosts that you can use to quickly deploy new hosts with the same configuration settings as other hosts in your environment. It can also be used to ensure configuration settings are current and up to date. Host profiles are attached to a host, and vCenter then ensures the host is in compliance with the host profile settings. If a host is not in compliance, it can be remediated, bringing the settings back to the standards you have defined in your host profile.

You typically have a host profile to match a specific server build. For example, if you use Cisco UCS blade servers, you might have one host profile for blades with local storage and another host profile for blades that have no local storage. A host profile expecting local storage would report a host with no local storage as noncompliant.

Host profiles are created from an existing host, which is already configured with all of the configuration settings you want. Once the host profile is extracted from the reference host, you can attach the host profile to other hosts.

For more information on host profiles, see the vSphere host profiles documentation at `http://pubs.vmware.com/vsphere-55/topic/com.vmware.ICbase/PDF/vsphere-esxi-vcenter-server-55-host-profiles-guide.pdf`.

Auto deploy

Auto deploy is a provisioning option for deploying ESXi hosts at boot time. There are two options when using auto deploy:

- **Stateless**: ESXi hosts use a **Preboot Execution Environment** (**PXE**) to request an image from the auto deploy server. Auto deploy pushes an ESXi image into the host's memory and runs ESXi in the memory as opposed to installing ESXi on the disk.
- **Stateful**: This feature installs ESXi over the network to a local disk in the server. Subsequent reboots of the server run from the local disk where ESXi was installed.

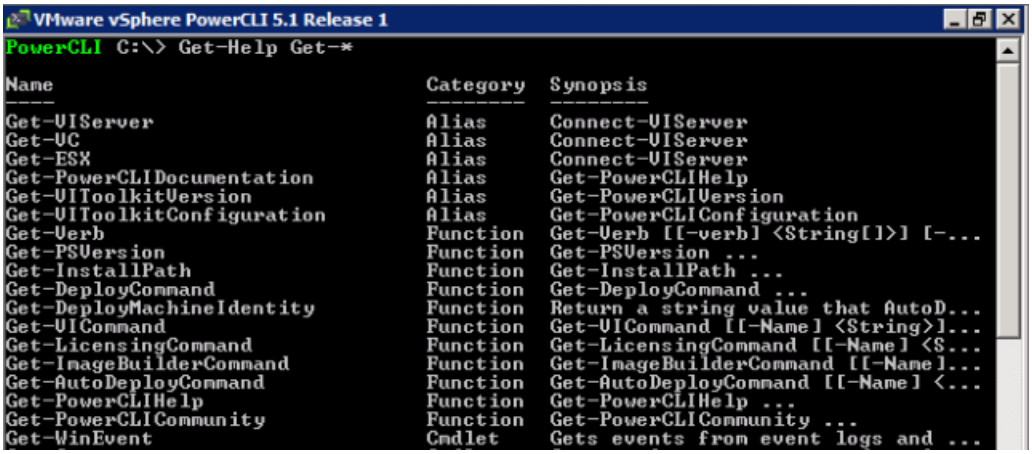
Images are created using the vSphere ESXi image builder, a set of PowerCLI cmdlets used to create the image add third-party drivers to support your host. In addition, auto deploy relies heavily on host profiles because every time the server is restarted, ESXi is installed into memory. All configuration settings such as virtual switches, logfile settings, and other configurations are supplied through the host profile.

Auto deploy tools, such as host profiles, require vSphere Enterprise Plus. Complete documentation can be found in the VMware vSphere 5.5 Documentation Center at `http://pubs.vmware.com/vsphere-55/index.jsp#com.vmware.vsphere.install.doc/GUID-D0A72192-ED00-4A5D-970F-E44B1ED586C7.html`.

PowerCLI basics

PowerCLI is a set of Windows PowerShell cmdlets; as such, you need to install PowerCLI on a Windows-based computer with PowerShell 2.0 or 3.0 and .NET 2.0 SP2 or higher. Once PowerCLI is installed, you can start using it in place of the vSphere web client. The most useful and important command to remember is `Get-Help`. The `Get-Help` command is used to get help on a specific command or even find commands. After `Get-Help`, the two most common types of commands are `Get-` and `Set-`. The `Get-` command will return some type of information and `Set-` will alter the configuration to the specified setting.

Say you want to get the name of the ESXi hosts connected to vCenter, but you are not sure of the command to do so. You could correctly assume that there is most likely a `Get-` command related to that. If you type `Get-Help Get-*`, it will return a list of all the `Get-` commands.



Scrolling through the list, you will see a command called `Get-VMHost`. Typing that command will return a list of all hosts connected to vCenter along with its power state, number of CPUs, and CPU usage. Compare that with logging into the vSphere web client, clicking on **Hosts and Clusters**, clicking on your cluster, clicking on the **Related Objects** tab, and clicking on the **Hosts** tab. Typing `Get-VMHost` is much quicker.

Also, using the published VMware vSphere 5.5 documentation would be much quicker than reading through all of the returned commands. A list of all PowerCLI cmdlets can be found at `http://pubs.vmware.com/vsphere-55/index.jsp#nav/2_2_1_0`.

Now, let's consider a request to change the amount of memory in a VM from 4 GB to 8 GB (memory hot plug is supported). By using a `Set-` command, in this case `Set-VM`, you can change the amount of memory on your VM through one command:

```
Set-VM -VM <name_of_vm> -MemoryMB 8192
```

This is pretty quick compared to logging into the vSphere web client, clicking on VMs and templates, finding your VM, right-clicking on the VM, clicking on **Edit Settings**, changing the amount of memory, and clicking on the **OK** button.

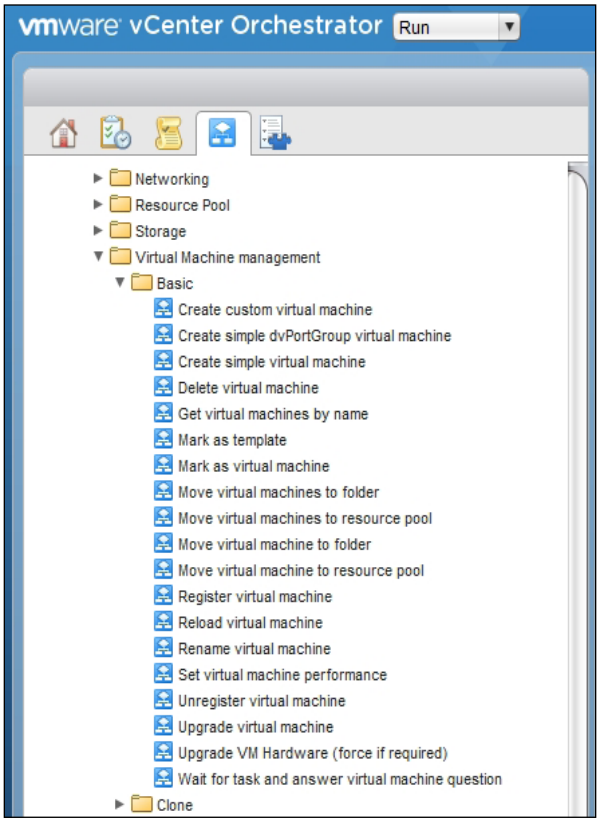
As you can see, knowing just a few common PowerCLI cmdlets can save quite a bit of clicking around in the vSphere web client. What is really cool about PowerCLI though is its ability to take multiple commands and create reusable scripts.

vCenter Orchestrator basics

vCenter Orchestrator (**vCO**) is a tool for developing advanced workflows within your environment, and it provides thousands of sample workflows, which you can start leveraging right away to help you automate your environment. vCO could even be used as a self-service portal for your organization as it maintains a separate access control list outside vCenter.

vCO is deployed with vCenter on Windows or as a separate appliance-based VM. Once installed, you can access vCO within vCenter or through its own client, which supports Windows, OSX, or Linux.

Out of the box, vCO provides thousands of workflows you can start using right away such as creating a new VM, deleting a VM, or cloning other VMs, as seen in the following screenshot:



The sample workflows provided are not editable; however, you can clone any existing workflow and edit it to meet your specific use case requirements.

With the out of the box workflows available with any licensed version of vSphere, there is really no reason to delay on automating routine tasks in your environment.

Automating resource management

In this section, we will look at how we can use PowerCLI and vCO to create a new VM. You will see how these automation tools can make these tasks easier and more consistent.

Creating a new VM

Creating new VMs is one of the most common tasks in vSphere and following the new VM wizard is quite easy. Let's look at an example request you may receive.

As the department responsible for your organizations in the vSphere environment, your team is asked to create new VMs for the development team. All development VMs must be created on the host specified in the request. They are in a folder named `development` for Windows Server 2012 with 2 vCPU, 8 GB of memory, a 40 GB of thin provisioned disk on the requested datastore. The Virtual Machine hardware version must be set to version 9, and add two VMXNET3 virtual NICs that are connected to the development port group and the VM Network port group.

For something as basic as creating a new VM, that's quite a list of requirements. If we were to do this simple task manually, you would do the following:

1. Log in to the vSphere web client.
2. Click on **VMs and Templates**.
3. Click on **Create a new virtual machine**.
4. Select **Create new virtual machine** and click on **Next**
5. Name the virtual machine, select the appropriate folder, and click on **Next**.
6. Select the appropriate host and click on **Next**.
7. Select the datastore cluster and click on **Next**.
8. Select the requested compatibility mode and click on **Next**.
9. Select the requested guest OS type and click on **Next**.
10. Conform or change the following options:
    - The number of vCPU to `2`
    - The amount of memory to `8`
    - The hard disk size to `40`; expand **New Hard disk** and select **Thin provision**
    - Set the first virtual NIC to the development port group, expand **New Network**, and change the adapter type to `VMXNET3`
    - Click on the pull-down option for **New device**, select the network, and click on the **Add** button
    - Change the new virtual NIC to the VM network port group, expand **New Network**, and change the adapter type to `VMXNET3`
    - Click on **Next**
11. Confirm your settings and click on **Finish**.

That's about 33 mouse clicks, or as I like to think of it, 33 chances to get the request wrong. Thanks to the features and tools provided by VMware, we can cut down on all of those manual mouse clicks and typing.

Creating a new VM with PowerCLI

By leveraging PowerCLI, we can eliminate those manual steps and combine them into a single command as follows:

```
New-VM -Name dev-vm -Location development -VMHost esxi-host -Datastore
esxi-host-datastore -Version v9 -GuestId windows8Server64Guest -NumCpu
2 -MemoryMB 8192 -DiskMB 40960 -DiskStorageFormat Thin -NetworkName
development, "VM Network"
```

By entering a single command, we were able to complete what 33 mouse clicks and some typing would have required. You might notice, however, a couple of problems with the preceding command.

Firstly, we did not specify the adapter type of VMXNET3, that here, one of the requirements for creating a new VM, because the `New-VM` cmdlet does not accept such a parameter. One of the great things about PowerCLI (and PowerShell for that matter) is its ability to pipe to other commands. Since we cannot change the network adapter with the `New-VM` cmdlet, we simply pipe to other cmdlets to achieve the requirement:

```
New-VM -Name dev-vm -Location development -VMHost esxi-host -Datastore
esxi-host-datastore -Version v9 -GuestId windows8Server64Guest -NumCpu
2 -MemoryMB 8192 -DiskMB 40960 -DiskStorageFormat Thin -NetworkName
development, "VM Network" | Get-NetworAdapter | Set-NetworkAdamter -Type
vmxnet3
```

Now, when the `New-VM` cmdlet finishes, the `Get-NetworkAdapter` cmdlet will retrieve the network adapters and set them to `VMXNET3` using the `Set-NetworkAdapter` cmdlet.

While the preceding command is useful, there are still a couple more problems. We have the name of the VM embedded within the command; once we create the VM, we cannot create another VM with the same name. The name of the ESXi host is also embedded, again not ideal as future requests might request an alternate host. Have no fear, PowerCLI's power also goes beyond using multiple cmdlets, and it also has the ability to combine multiple cmdlets into a script to achieve exactly what we want. Let's take a look at a script that meets all the requirements we described and can be reused because it will prompt for information that is likely to change from request to request:

```
$VMName = Read-host "Enter the name of the VM you wish to create"
```

```
#Lists host for the connected vCenter, prompts user to enter the desired
host location for the VM

Get-VMHost | Format-Wide

$HOSTName = Read-host "Enter the Name of the Host you wish to create the
VM on"

#Lists datastores for the connected vCenter, prompts user to enter the
desired datastore location for the VM

Get-Datastore | Format-Wide

$DSName = Read-Host "Enter the name of the datastore you wish to create
the VM on"

#Lists VM folders for the connected vCenter, prompts user to enter the
desired folder location for the VM

Get-Folder | Format-Wide

$FOLDERName = Read-host "Enter the name of the folder you wish to create
the VM in"

#Takes input from above steps and passes them to the New-VM command for
VM creation

New-VM -Name $VMName -Location $FOLDER Name -VMHost $HOSTName -Datastore
$DSName -Version v9 -GuestId windows8Server64Guest -NumCpu 2 -MemoryMB
8192 -DiskMB 40960 -DiskStorageFormat Thin -NetworkName Development, "VM
Network"

#Once VM creation is completed, VM network adapters are converted to
vmxnet3

Get-VM $VMName | Get-NetworkAdapter | Set-NetworkAdapter -Type vmxnet3
```

You can see another example of piping to other cmdlets in the last line. We called the `Get-VM #$VMName` cmdlet (or you could type in a static name), then a pipe character (`|`) followed by another cmdlet. So long as the cmdlet can be piped (some cannot), you can string many of these together into a single command.

This example, while basic, uses several commands to display information about the environment, ask a user for input, perform an action, and make changes to an existing resource. You should also note that this example is not meant to reflect best practices, or even efficiency, rather it is a bare bones PowerCLI example of how multiple commands can be used within a script to perform common tasks.

You can find several sample scripts in the VMware vSphere Documentation Center at `http://pubs.vmware.com/vsphere-55/index.jsp#com.vmware.powercli.ug.doc/GUID-3E36F4EA-3742-48BA-BB4B-7E0A2EAAE83E.html`.

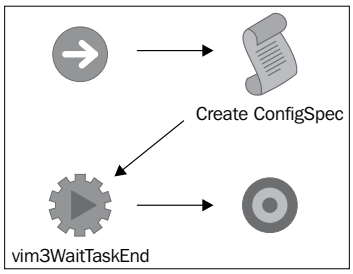
Creating a new VM with vCO

Let's look at what a vCO workflow looks like to create a new VM. There are two built-in workflows ready to go out of the box with vCO: **Create simple virtual machine** and **Create custom virtual machine**.

A workflow consists of several parts:

- **Inputs**: This is similar to what we used in our PowerCLI script to prompt for information
- **Outputs**: The expected outcome of the workflow, in this case, a new VM
- **Schema**: The steps used to move from start to finish

Here is the schema for the **Create simple virtual machine work flow**. Each of the icons in the workflow shown in the following figure represents an available action in vCO, which you can use to create your own workflows, though, there are many more than this.

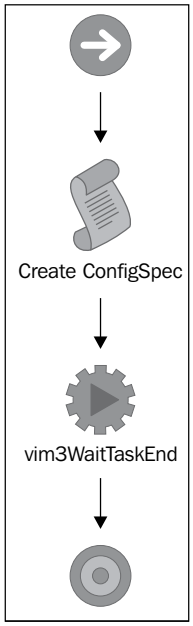


This workflow will look a bit different if you are following along with vCO at home so that we could fit the workflow into a smaller space. The task starts, collects information, runs through the process, and then ends. Now, let's run the workflow and see what it actually does.

1. Select the workflow in the vCO client and click on the green play button.
2. A form appears asking for the VM name, VM folder, size of the hard drive, memory, number of vCPU, guest OS type, and whether to thin provision.
3. Fill in all the information and click on **Next**. Note that if you click on a link and nothing appears, type something into the filter box. This is a quirk of vCO, one I hope they fix in a future release. For example, when you click in the Virtual machine guest OS box, an empty window appears. Type `Windows` in the filter box on the top-right corner, and it will display the matching items. Select your desired OS and click on the **Select** button.
4. Select the server, resource pool, network, and datastore and click on the **Submit** button.

The vCO workflow will take the input and create a new VM with the selected settings. If you have not created any resource pools in your environment, which is a required input of this workflow, expand **Hosts | Cluster** and you will see a pool called **Resources**, which you can select.

You can see in the following figure how the workflow takes place and creates my new VM.



As you can see, vCO can be quite powerful even if you are using out of the box workflows. In addition to the default workflows available, or those that you can create, VMware has created a site at `https://solutionexchange.vmware.com` where you can download additional workflows. The VMware vSphere documentation center, as you might expect, has information on installing, creating, and securing vCO at `http://pubs.vmware.com/vsphere-55/index.jsp`.

Community automation resources

While the technology that VMware vSphere provides is excellent, one of the things that really sets it apart is not only the technology, but the communities they have helped to build it. These are people helping one another through the official VMware communities forum, individual blogs, and other social media outlets. Here are some great free examples and community resources related to automation:

VMware education services (`www.vmwarelearning.com`) is a website for free instructional videos on many virtualization topics, which include videos on vCenter Orchestrator.

VMware VMTN community refers to the official VMware community sites that include the vCO and PowerCLI communities, which are available at the following links:

- `https://communities.vmware.com/welcome`
- `https://communities.vmware.com/community/vmtn/vcenter/orchestrator`
- `https://communities.vmware.com/community/vmtn/automationtools/powercli`

Above and beyond the official VMware resources, there are several great user-contributed sites as well.

Alan Renouf (`https://twitter.com/alanrenouf`) publishes several useful PowerCLI scripts at `http://www.virtu-al.net`. He also publishes one-liners, which are short and typically piped PowerCLI commands to perform specific tasks (`http://www.virtu-al.net/2009/07/07/powercli-more-one-liner-power/`). For example, he has a one liner that will help you determine which VM has a specific MAC address; this is useful if you are performing any network troubleshooting:

```
Get-vm | Select Name, @{N="Network";E={$_ | Get-networkAdapter | ? {$_.
macaddress -eq "00:50:56:A1:50:43"}}} |Where {$_.Network-ne ""}
```

Matt Vogt (`https://twitter.com/mattvogt`), who blogs at `http://blog.mattvogt.net`, publishes articles on various virtualization topics. What makes the VMware community great is that you are likely to find others working on similar projects or who have run into similar problems. Matt published a blog post called *VMs Grayed Out After NFS Datastore Restored*, which was a modified PowerCLI script from another VMware community member Raphale Schitz. It will search vCenter for VMs that are marked as inaccessible and reconnect them.

This particular script saved me quite a bit of time after performing maintenance on a NAS servicing one of my vSphere clusters. I needed to shut down all of the VMs to carry out this maintenance; however, due to the time gap, the NAS was disconnected, the VMs were not able to reconnect, something I had not accounted for. This script is a great example of how scripting and automation can save time and how helpful the members of the VMware community are. You can find this script at `http://blog.mattvogt.net/2013/02/08/vms-grayed-out-after-nfs-datastore-restored/`.

If you are looking for some community-contributed content on vCO, check out Mike Preston's blog at `http://blog.mwpreston.net`, where he blogs on various virtualization topics. In addition to vCO-related posts at `http://blog.mwpreston.net/tag/vco/`, he has also recently published a series on the VCAP-DCA exam where he reviews lesser known features of vSphere that are part of the blueprint and has a great VCP study guide.

The last site featured here, which we have referenced before, is `http://professionalvmware.com`, a group of community members who host podcasts on various virtualization topics. There were a series of podcasts dedicated to automation to wrap up 2013, which included PowerCLI, vCloud Automation Center, and vCO. You can find the podcasts at `http://professionalvmware.com/vbrownbag-automate-all-the-things-training-schedule/` or on their official iTunes channel.

This is certainly not all of the community-contributed content. You should make it a point to get involved with the virtualization community as it provides a great opportunity to learn and connect with others.

Available monitoring options

There are plenty of monitoring tools in the market; the need to monitor your servers and infrastructure was needed well before virtualization. However, VMware does have tools that can help you track the health of your environment and even help with forecasting.

In the following sections, we will look at the application/feature mentioned in the following table:

Application/feature	Description	Licensed with
Alarms	This provides the ability to monitor hosts and VMs for certain alerts.	This is included in all licensed editions of vSphere.
vCenter Operations Manager	This provides the ability to monitor hosts, VMs and datastores and providing trending and analysis	This product is including the vCenter Operations Manager Suite or as a separate product

Alarms

Alarms are a basic but useful way to provide alerts on specific environment conditions in vCenter. Alarms can provide us with alerts on both VM-and host-level issues, including configuration problems. Since they are included in vCenter, there is generally little reason for not configuring some alarms.

> You should avoid over-monitoring or having too many of the same processes monitored so you are not overwhelmed by alert e-mails.

Alarms maintain three states: green, yellow, and red. Compare this with Nagios, a popular open source monitoring application that would be the equivalent of *OK*, *warning*, and *critical*. After vCenter, the only requirement for alarms is to set up SMTP, assuming you want e-mail alerts. You could set up alarms with no SMTP server configured, but you would need to log in to the vSphere web client to see the alarms and might miss something important.

Assuming you have an accessible SMTP server, all you need to do to configure SMTP is log in to the vSphere web client, navigate to **vCenter | vCenter Servers**, select your vCenter server, click on the **Manage** tab, and then click on the **Edit** button. Click on **Mail** and fill in the **Mail server** text with the address of your SMTP server and **Mail sender** textbox with the *from* address.

You can see the alarm definitions available under the **Manage** tab for any object managed by vCenter. Although some, such as alarms for VMs, may have fewer alarms than a host. You wouldn't configure an alarm to monitor a host failure on a VM, for example.

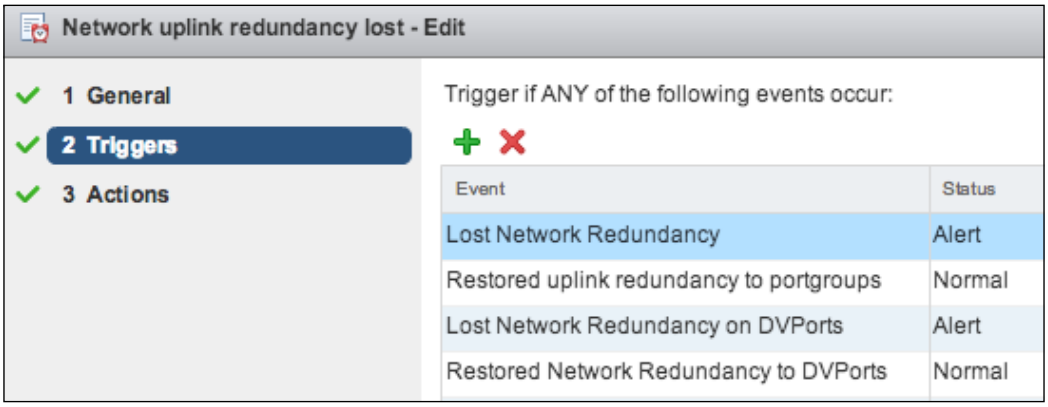
Let's configure an alarm to notify us if one of our vSwitches does not have a redundant uplink (something that might be hard to monitor in third-party monitoring tools).

Configuring an alarm

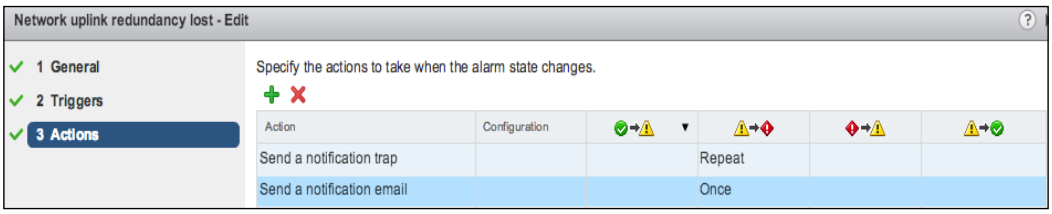
Configuring alarms is very straightforward and should be done in most situations.

1. Log in to the vSphere Web Client
2. Navigate to **vCenter | vCenter Servers** and select the vCenter server you wish to configure alerts on.
3. Click on the **Manage** tab and then click on **Alarm Definitions**.
4. Select **Network uplink redundancy lost** and click on the **Edit** button.

5. On the **General** screen, you can only edit a few items; however, in this case, you will most likely want to leave it to the host to monitor the items so you can monitor a specific event, which we will configure next. Click on the **Next** button.
6. By default, this alarm ships with a few options that are preconfigured. Refer to the following screenshot; it will trigger an alert for **Lost Network Redundancy** and **Lost Network Redundancy on DVPorts**.



7. Here, you could add additional **Triggers** if you wish:
   - Click on the green **+** icon and a new field will be added to the list of events.
   - Click on the pull-down menu to scroll through the available events and select **Network Redundancy Degraded**, which ensures the **Status** is set to **Alert**.
8. Whether you add additional triggers or use defaults, click on the **Next** button.
9. The **Actions** screen is where you can configure how you are notified. You'll notice four columns for the options: *OK* to *warning*, *warning* to *critical*, *critical* to *warning*, and *warning* to *OK*.



10. Click on the green **+** icon, and it will add the **Send a notification email** option. Click on the field for which you wish to be notified. For example, if you want an e-mail notification when it goes from *OK* to *warning*, click on that field and select either **Once** or **Repeat** and click on **Finish.**

You can also manage alarms through PowerCLI; there is a great blog post on the available cmdlets at `http://blogs.vmware.com/vipowershell/2010/12/managing-vsphere-alarms-with-powercli.html`.

This is just one example of alarms; there are many you can configure to keep you informed about your environment. Planning which alarms you can use via vCenter and which you can manage through other monitoring tools requires some thought so you don't receive multiple alerts from a single event.

vCenter Operations Manager

vCenter Operations Manager, commonly referred to as vCOPS, is an advanced monitoring tool for vSphere environments. vCOPS is available in several editions. If you are licensed for vCenter, then at the very least, you have access to the vCOPS foundation edition. This includes monitoring, alerts, and self-learning analytics that understand how your environment functions. To understand how self-learning analytics work, let's consider a basic example.

Every night you run backups (or you should!), and the VM running on your backup software will typically have increased CPU utilization during the backup window, which is expected and normal for that workload. A typical monitoring application would not be aware of this behavior on its own; you would have to manually configure an exception or maintain a window where the software would not trigger an alert. The self-learning alerts in vCOPS would understand that this is normal and would not trigger an alert during the time window where this increase occurs.

vCOPS is deployed as an OVA (virtual appliance); there are just a few requirements. During the OVA deployment, you will be asked which server to deploy to, as you might expect; however, you won't be able to proceed without enabling DRS. If you are not licensed for Enterprise Plus, you will need to remove a host from the cluster. In order to deploy the OVA, you need to have a standalone host (refer to `http://kb.vmware.com/selfservice/microsites/search.do?language=en_US&cmd=displayKC&externalId=2013695`).

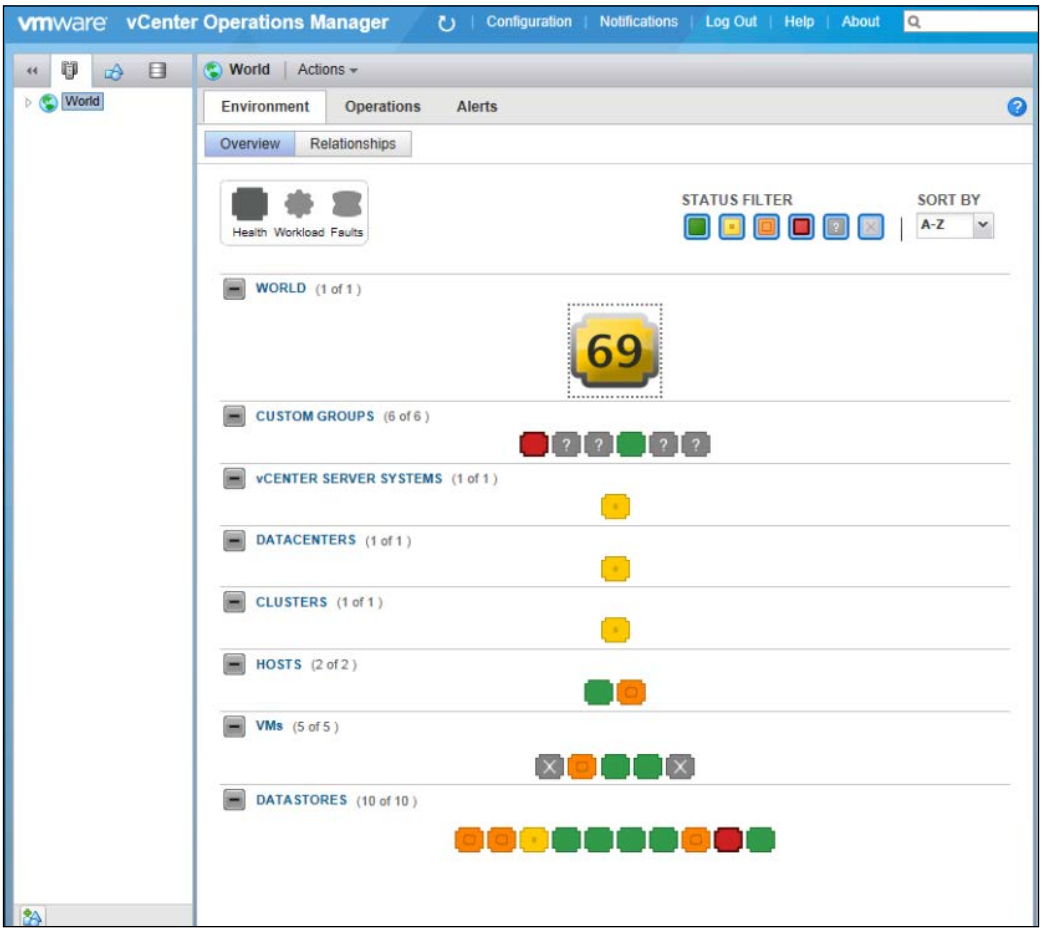
You will be prompted to select a configuration size. Each of these settings defines how many resources will be assigned to the VMs so that they can support your environment.

- **Small (less than 1500 VMs)**: 4 vCPU and 16 GB of RAM

- **Medium (1500 to 3000 VMs)**: 8 vCPU and 25 GB of RAM
- **Large (greater than 3000 VMs)**: 16 vCPU and 24 GB of RAM

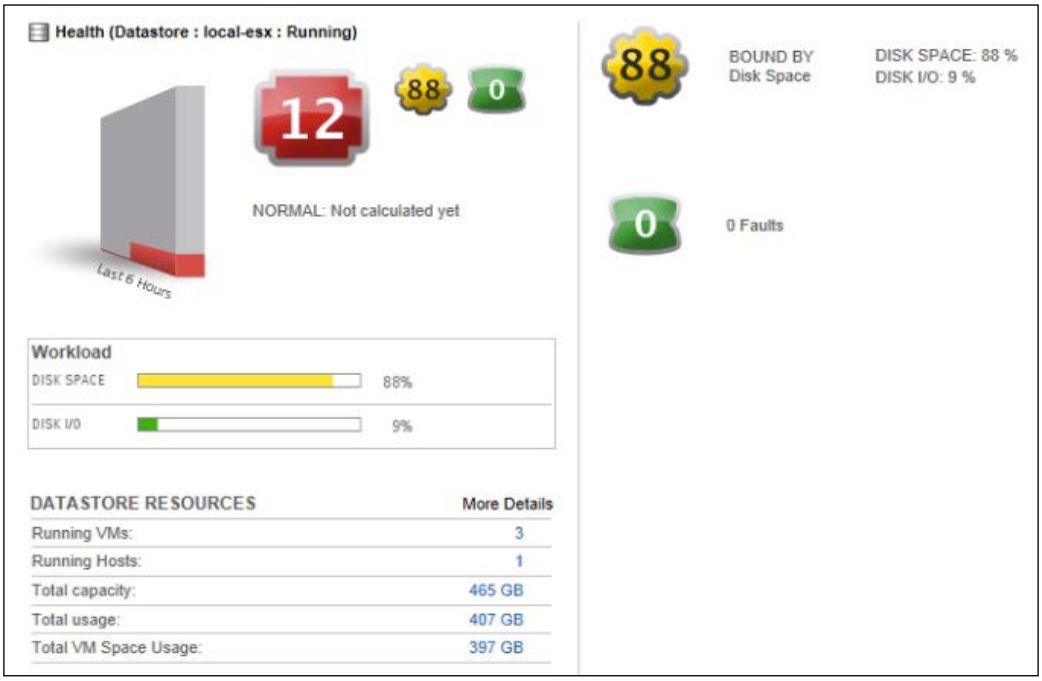
You can download the OVA from `my.vmware.com` and deploy any appliance in your environment. You can find full details on the installation at `https://www.vmware.com/pdf/vcops-vapp-57-deploy-guide.pdf` or a quick start install guide I published at `http://www.virtxpert.com/deploying-vcenter-operations-manager-appliance/`.

Once you have vCOPS installed and configured, you are able to view the health of your environment very easily. vCOPS factors all of the various parts of your environment into a world and provides a score based on the various factors in your environment. As you can see in the following screenshot, I have a score of 69, which vCOPS classifies as *abnormal* versus *good* (green) or *degraded* (orange).



As you can see from the preceding screenshot, vCOPS further breaks down the various components that make up the **WORLD**. You can quickly see that one of my custom groups is in a *bad* (red) state and that several of my datastores are *degraded* and only one is *bad*.

If you double-click on the red icon in the Datastores section, it will bring you to the associated datastore and give you an overview of its health. As you can see here, my datastore was classified as bad likely because I have very little free disk space; otherwise, the I/O on that datastore is fine.



Summary

In this chapter, we looked at how automation can reduce the amount of effort to manage your VMware vSphere environment with the tools that VMware makes available: vCenter Orchestrator and PowerCLI. We also reviewed two monitoring options: alarms, which are built into vCenter, and vCenter operations manager, which comes in both a free foundation edition and advanced editions that require additional licensing.

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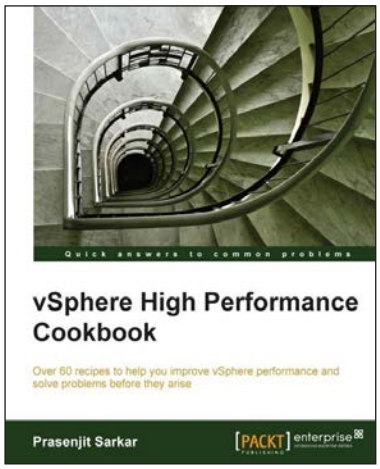

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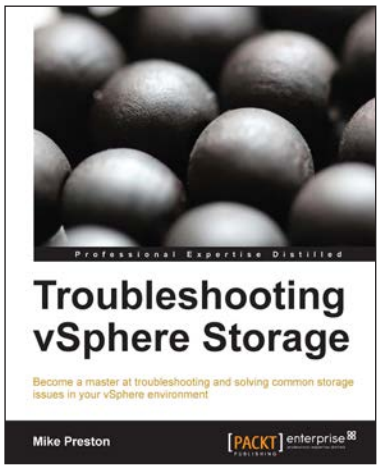

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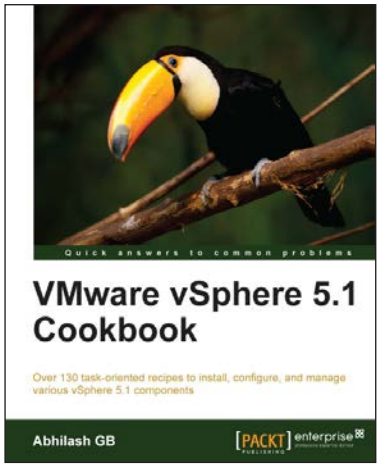

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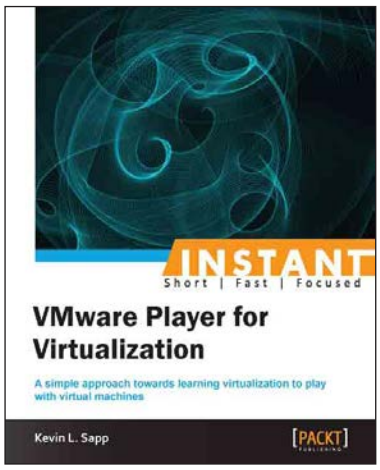

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